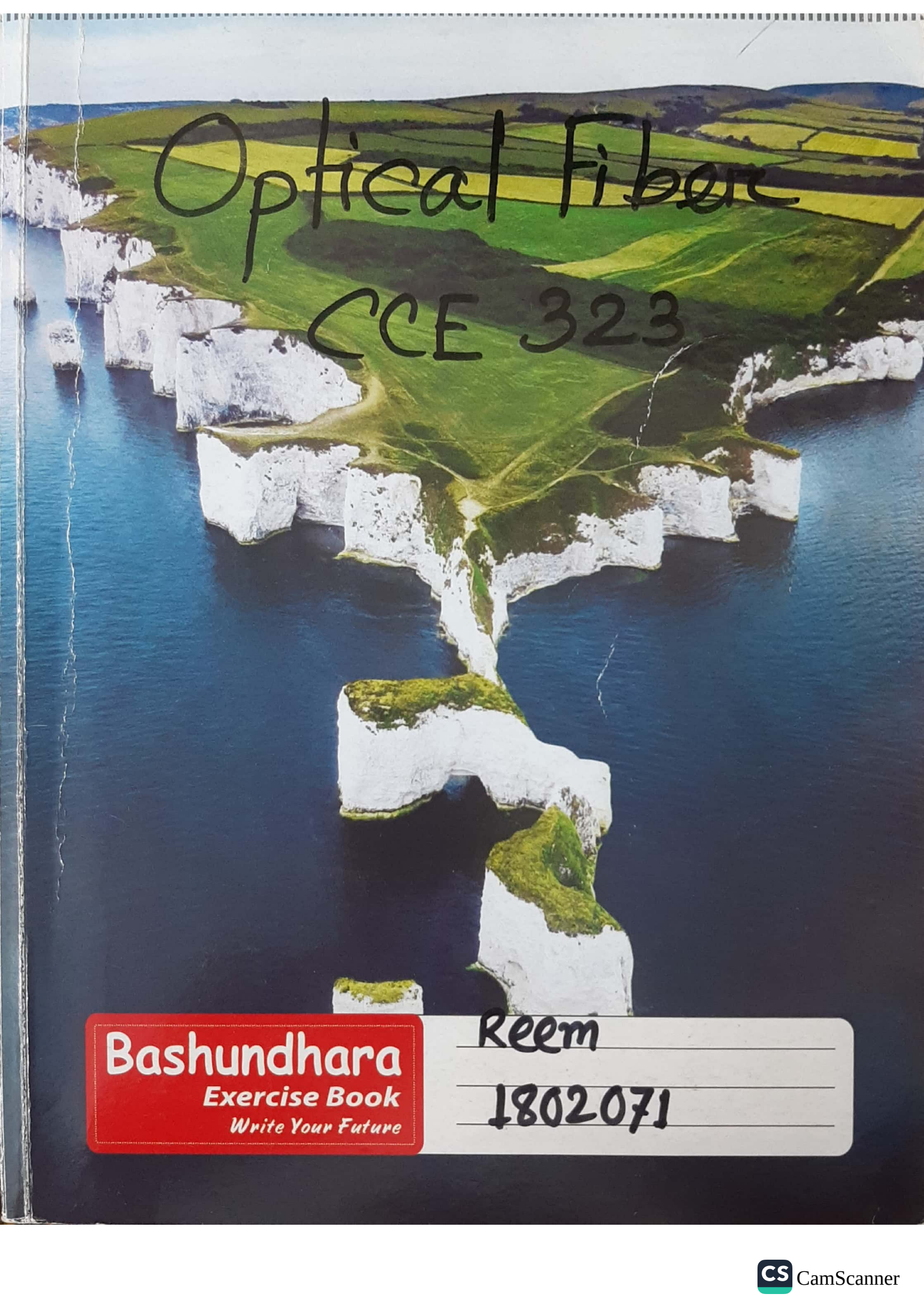




Optical Fiber

CCE 323

Bashundhara

Exercise Book

Write Your Future

Reem

1802071

Reference Books

1. Optical fiber Communication - Principles and Practice
3rd Edition - John M. Senior
2. Optical Fiber Communications
Gerard Keiser

CCE 323 Optical Fiber Communication

Chapter 04

Optical Fibers and Cables

Introduction

Preparation of optical fibers

Liquid phase techniques

Fiber drawing

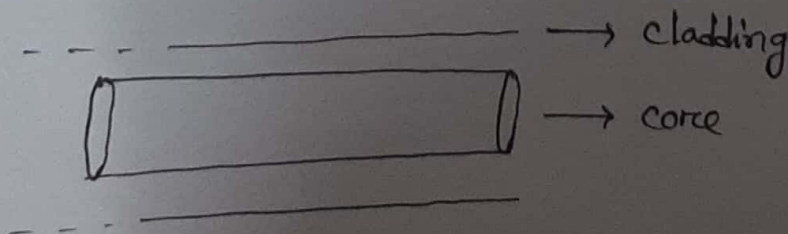
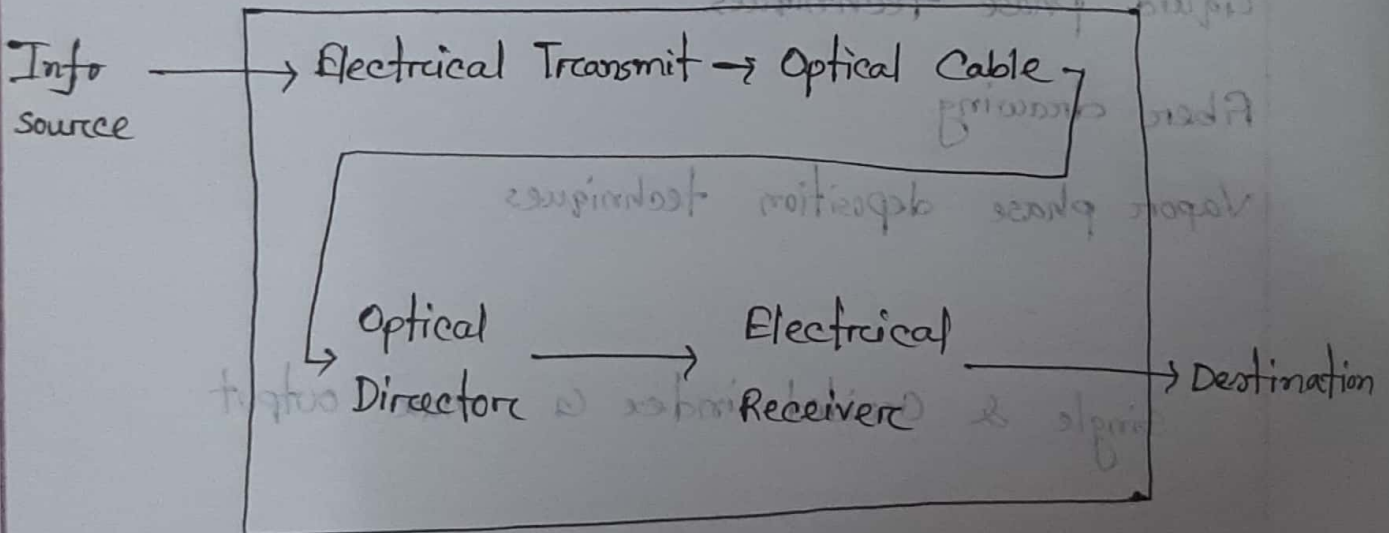
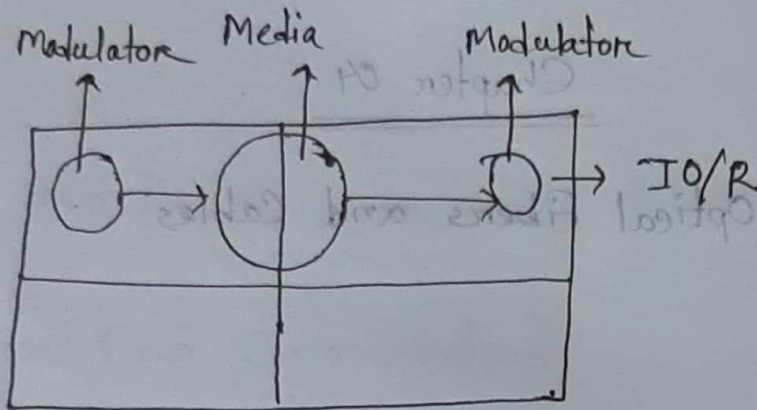
Vapor phase deposition techniques

Single & Graded index & max output



Chapter 01

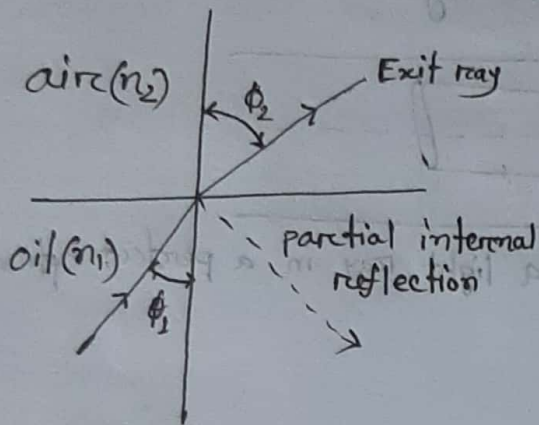
II



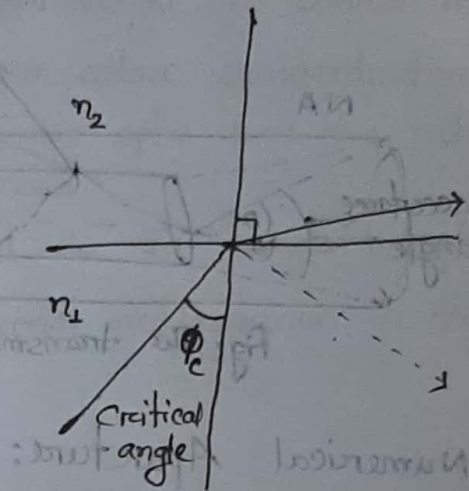
Core এর reflective index হলি cladding এর থেকে। Core
মত হলি pure গর reflective index হলি

Chapter 02

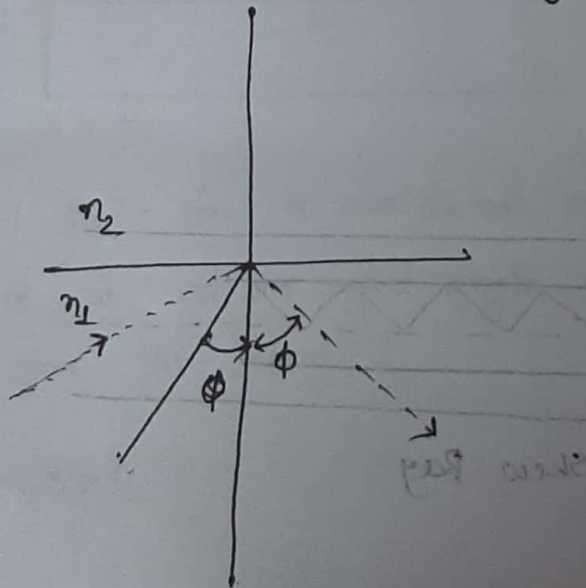
* Ray Theory Transmission



(a)
refraction



(b)
the limiting case of refraction
showing the critical ray at an
angle ϕ_c



(c) total internal reflection where $\phi < \phi_c$

$$n_1 \sin \phi_1 = n_2 \sin \phi_2$$

$$\Rightarrow \frac{\sin \phi_1}{\sin \phi_2} = \frac{n_2}{n_1}$$

$$\Rightarrow \sin \phi_c = \frac{n_2}{n_1}$$

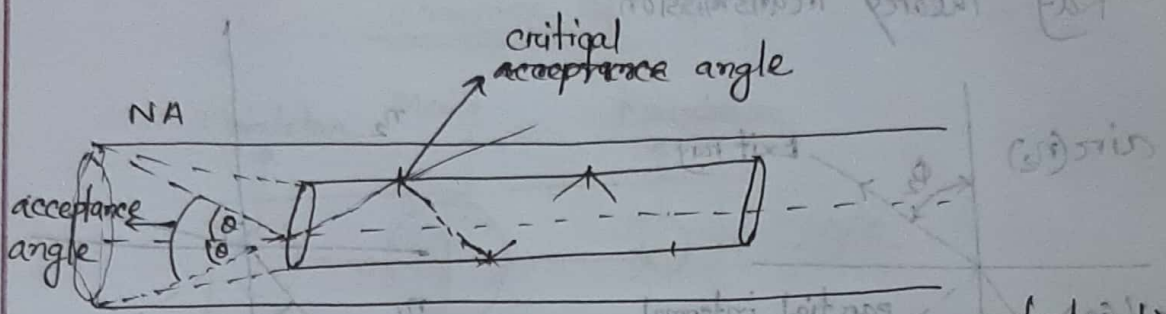
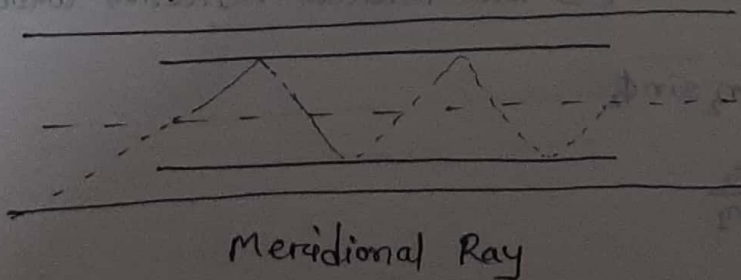
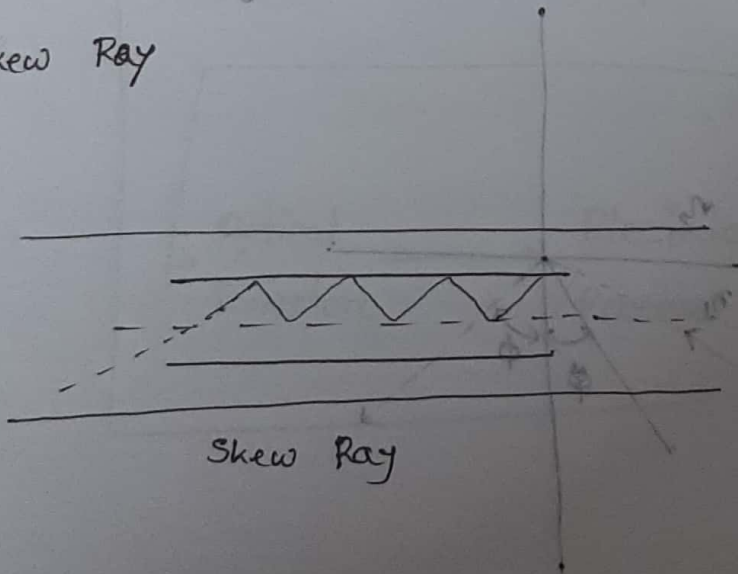


fig: The transmission of a light ray in a perfect optical fiber

Numerical Aperture:

* Meridional Ray

* Skew Ray



Acceptance Angle: is the maximum angle at which incoming sunlight can be captured by a solar concentrator. Its value depends on the concentration of the optic and the refractive index in which the receiver is immersed.

Numerical Aperture: is defined as the largest angle an incident ray can have for total internal reflectance in the core.

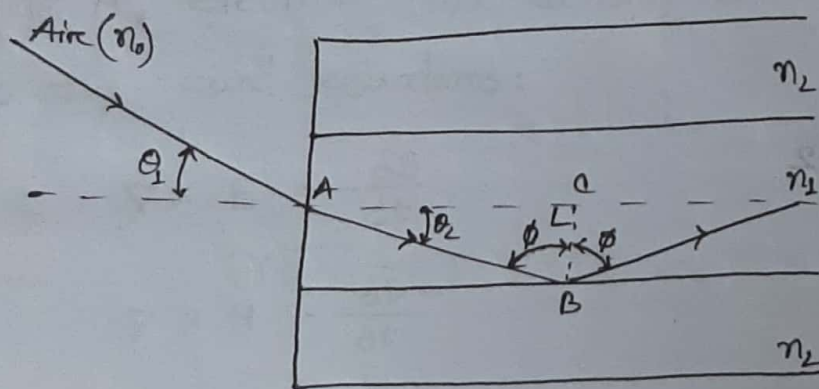


Fig: The ray path for a meridional ray launched into an optical fiber in air at an input angle less than the acceptance angle for the fiber

$$n_0 \sin \theta_1 = n_1 \sin \theta_2$$

$$\phi = \frac{\pi}{2} - \theta_2 \quad \text{from ABC triangle}$$

$$\Rightarrow n_0 \sin \theta_1 = n_1 \cos \phi$$

$$\Rightarrow n_0 \sin \theta_1 = n_1 (1 - \sin^2 \phi)^{\frac{1}{2}} \quad [\sin^2 \phi + \cos^2 \phi = 1]$$

$$\Rightarrow n_0 \sin \theta_a = (n_1^2 - n_2^2)^{\frac{1}{2}} \quad [\phi \text{ becomes equal to the core cladding critical angle}]$$

$$\Rightarrow NA = n_0 \sin \theta_a = (n_1^2 - n_2^2)^{\frac{1}{2}}$$

$\Delta n = n_1 - n_2$; NA in terms of relative refractive index difference

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

$$\approx \frac{n_1 - n_2}{n_1} \text{ for } \Delta \ll 1$$

$$NA = n_1 (2\Delta)^{\frac{1}{2}}$$

Example 2.1, 2.2

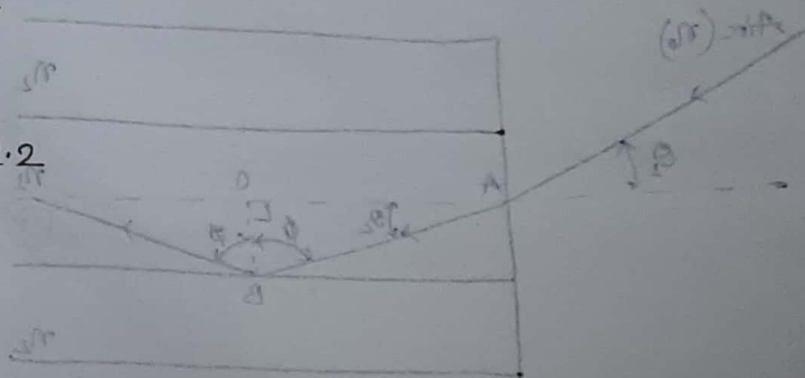


Fig: The ray path for a meridional ray launched into an optical fiber in air at an input angle less than the acceptance angle for the fiber

$$n_0 \sin \theta_0 = n_1 \sin \theta_1$$

$$\phi = \frac{\pi}{2} - \theta_1 \text{ from ABC triangle}$$

$$\sin \theta_1 = \cos \phi$$

$$n_0 \sin \theta_0 = n_1 \cos \phi$$

$$n_0^2 \sin^2 \theta_0 = n_1^2 \cos^2 \phi$$

$$n_0^2 \sin^2 \theta_0 = n_1^2 (1 - \sin^2 \phi)$$

* Electromagnetic mode theory for optical propagation

Electromagnetic waves propagation is provided by Maxwell's equation.

A medium with zero conductivity these vector relationships may be written in terms of the electric field E , magnetic field H , electric flux density D and magnetic flux density B as the curl equations:

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \times H = \frac{\partial D}{\partial t}$$

divergence equations

$$\nabla \cdot D = 0 \quad (\text{no free charges})$$

$$\nabla \cdot B = 0 \quad (\text{no free poles})$$

where ∇ is a vector operator

The four field vectors are related by the relations:

$$D = \epsilon E$$

$$B = \mu H$$

where ϵ = dielectric permittivity

μ = magnetic permeability

Substituting D and B and taking the curls of eqn's

① and ② $\Rightarrow$

$$\nabla \times (\nabla \times E) = -\mu \epsilon \frac{\partial^2 E}{\partial t^2}$$

$$\nabla \times (\nabla \times H) = -\mu \epsilon \frac{\partial^2 H}{\partial t^2}$$

Using the divergence conditions of Eqn ③ and ④ with the vector identity $\Rightarrow$

$$\nabla \times (\nabla \times Y) = \nabla (\nabla \cdot Y) - \nabla^2 Y$$

We obtain the nondispersive wave equations

$$\nabla^2 E = \mu \epsilon \frac{\partial^2 E}{\partial t^2}$$

$$\nabla^2 H = \mu \epsilon \frac{\partial^2 H}{\partial t^2}$$

where $\nabla^2 =$ Laplacian operator

$$\nabla^2 \psi = \frac{1}{v_p^2} \frac{\partial^2 \psi}{\partial t^2}$$

$$v_p = \frac{1}{(\mu \epsilon)^{\frac{1}{2}}} = \frac{1}{(\mu_r \mu_0 \epsilon_r \epsilon_0)^{\frac{1}{2}}}$$

$$c = \frac{1}{(\mu_0 \epsilon_0)^{\frac{1}{2}}}$$

$$\Rightarrow \nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \quad (\text{for cartesian co-ordinates})$$

or,

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial r^2} + \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{\partial^2 \psi}{\partial z^2} \quad (\text{for cylindrical polar coord})$$

$$\psi = \psi_0 \exp[j(\omega t - k \cdot r)]$$

$$k = \frac{2\pi}{\lambda}$$

* Modes in planar guide

Fig 2.8, 2.9

* Phase and Group velocity

Fig 2.10: Fiber splices: permanent or semi-permanent (conventional or electric soldered joint)

* Phase shift with total internal reflection and the evanescent field

* Goos-Haenchen shift

* Cylindrical fiber

modes: to couple all the light equally in separate

Fig. 2.15

mode coupling

Chapter 05

Fiber Joints and Couplers

* 2 types fiber-fiber joints:

i) Fiber splices: permanent or semipermanent joint
(analogous to electric soldered joints)

ii) Demountable fiber connectors or simple connectors:
these are removable joint (analogous to electrical
plugs & sockets)

Purpose: to couple all the lights ideally in one fiber
to ~~and~~ adjoint fiber

- to couple the fibers, the media has to be same
- Optical fiber communication takes place or occurs by joining or terminating the transmission media.
- Repeater is used to amplify the signal to cover long distance.
- Repeater overcomes communication loss
- no. of connection/joints depends on the link length between repeaters.

* Types of Joint coupling:

- i) Single - one to one (one fiber to another)
- ii) Single - one to many (one fiber to many fibers)
(split etc.)
- iii) Many to one (for bidirectional device)

* Losses in fiber joint:

1) Radiative loss: 2 types

- i) Macroscopic fiber bending
- ii) Microscopic fiber bending or Mode coupling

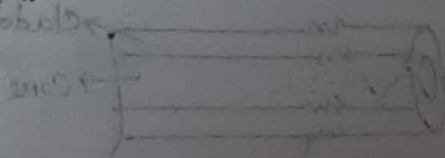
2) Misalignment loss: 3 types

- i) Core or longitudinal misalignment
- ii) Lateral misalignment
- iii) Angular misalignment

3) Gap losses

4) Losses due to difference in diameter

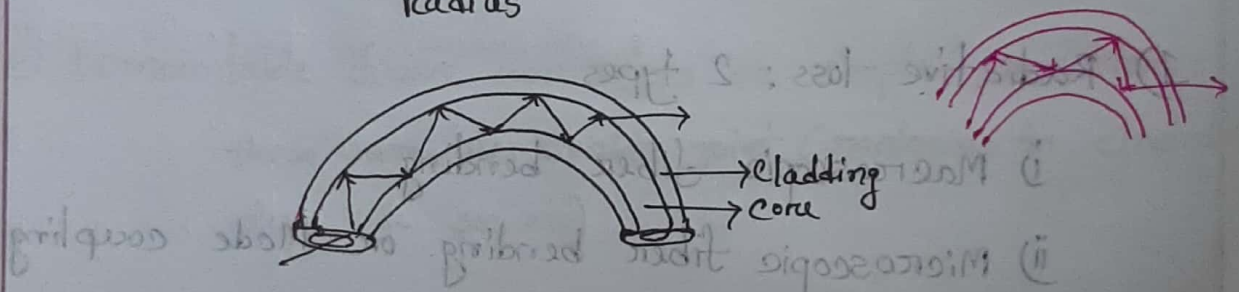
- 5) " " " change " Numerical aperture



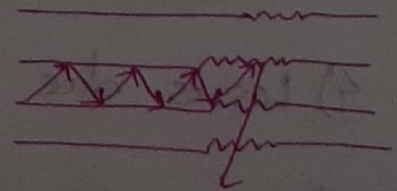
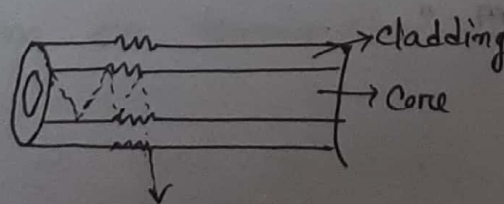
* Radiative loss: Instead of reflecting from the correct point, it reflects at certain distance.

- Reflective angle is created in the cladding
- Bending, breakage,

i) Macroscopic: For bending the wire, greater core radius

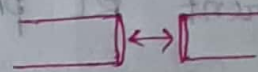
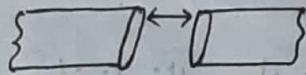


ii) Microscopic: Slight bend; cause light to be reflected at an angle where there is no further reflection.

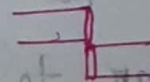
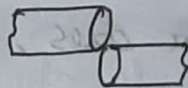


* Loss due to misalignment: The misalignment due to the fluid used for the air gap.

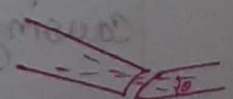
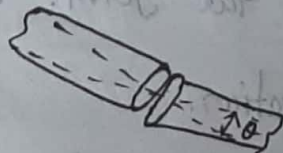
i) Longitudinal misalignment: separation between fibers



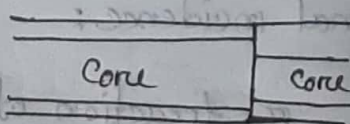
ii) Lateral misalignment: displacement of x, y axis; offset perpendicular to the fiber core axis



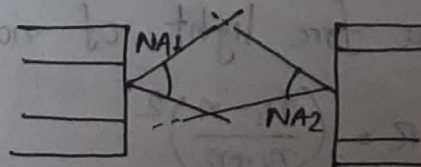
iii) Angular misalignment: angle between the core axis;



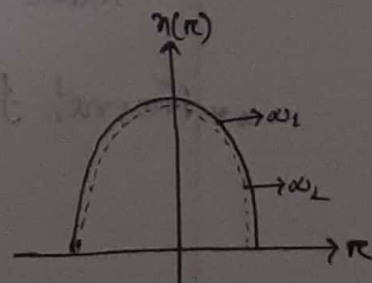
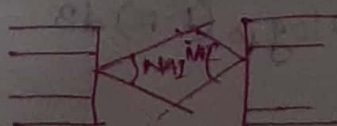
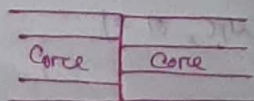
** Some intrinsic coupling losses:



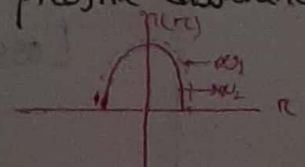
(a) Core diameter mismatch



(b) Numerical Aperture Mismatch



(c) refractive index profile difference



* Connection problems when jointing fiber with:

- i) Different core and/or cladding diameter
- ii) Different NA and/or relative RI differences
- iii) Different RI profiles
- iv) Fiber faults (core ellipticity, concentricity etc.)

* Loss due to Fresnel reflection:

Even when the two ^{joined} fiber ends are smooth and perpendicular to the fiber axes, and the two fiber axes are perfectly aligned, a small portion of the light may be reflected back into the transmitting fiber causing attenuation at the joint. This phenomenon is known as Fresnel reflection.

it is associated with the step changes in refractive index at the jointed interface.

* Fresnel formula for light of normal incidence:

$$r = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

r = fraction of light

n_1 = core RI

n_2 = medium RI

at $n_2 = 1$

* Loss due to Fresnel reflection:

$$\text{Loss}_{\text{Fres}} = -10 \log_{10} (1 - r) \text{ dB}$$

- reflection loss reduce ~~২০৪০ জয়~~
- To reduce reflection loss, index-matching fluid is used between the gaps.
- For this the 1st and 2nd interface gets separated. If they are of same magnitude, the loss is added after calculating separately.
- If fluid is not used, the equation is applied.

* Example 5.1: $n_1 = 1.5$, $n = 1$ (air)

Solⁿ: The magnitude of Fresnel reflection at the fiber-air interface is -

$$r = \left(\frac{n_1 - n}{n_1 + n} \right)^2 = \left(\frac{1.5 - 1}{1.5 + 1} \right)^2$$

$$= 0.04 = 4\% \text{ reflection}$$

Loss due to Fresnel reflection,

$$\text{Loss}_{\text{Fres}} = -10 \log_{10}(1 - r) = -10 \log_{10} 0.96$$

$$= 0.18 \text{ dB}$$

Optical loss at the second interface is also 0.18 dB

$\therefore$ Total loss = ~~0.38~~ 0.36 dB

* Insertion Loss Characteristics for joined fibers with misalignments:

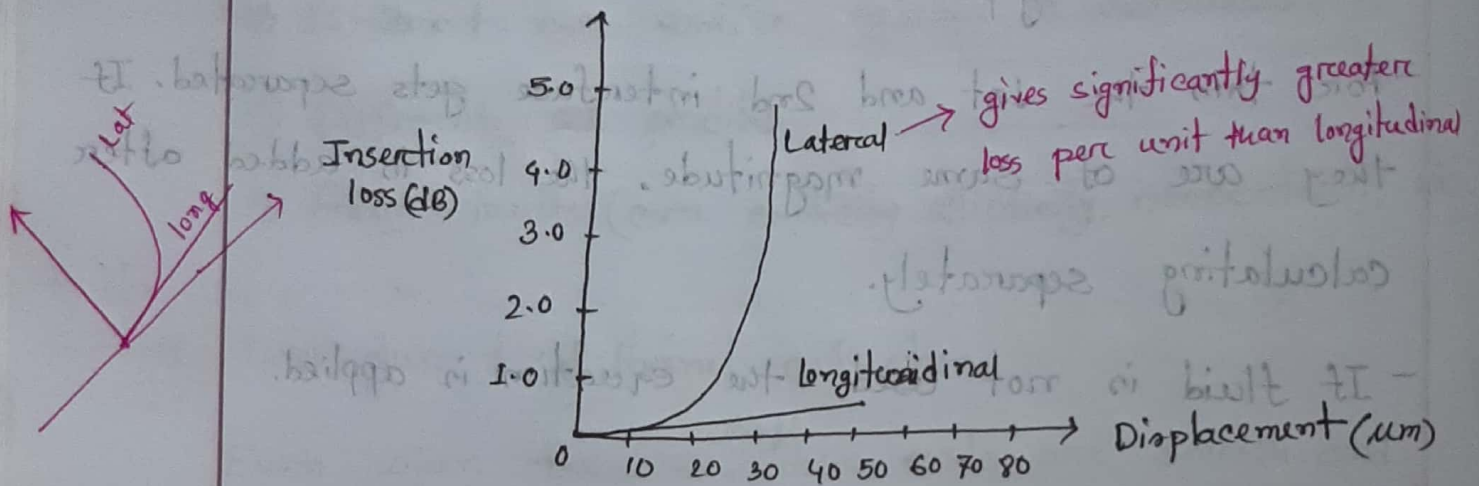


Fig: Insertion loss due to lateral & longitudinal misalignment

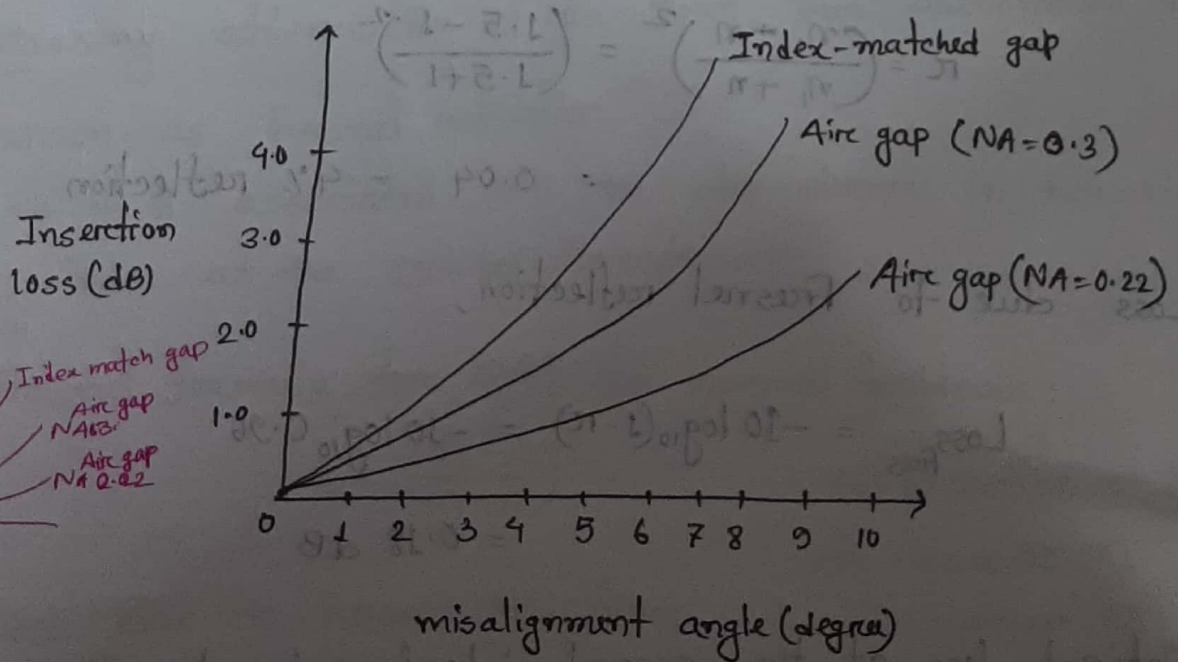


Fig: Insertion loss due to angular misalignment

* Multimode fiber joints

i) Lateral misalignment:

- reduces overlap region between two fiber cores
- Lateral coupling efficiency for two similar step index fibers:

$$\eta_{lat} = \frac{16(n_1/n)^2}{[1 + (n_1/n)]^4} \cdot \frac{1}{\pi} \left\{ 2 \cos^{-1} \left(\frac{y}{2a} \right) - \left(\frac{y}{a} \right) \left[1 - \left(\frac{y}{2a} \right)^2 \right]^{1/2} \right\}$$

$$\eta_{lat} = \frac{16(n_1/n)^2}{[1 + (n_1/n)]^4} \cdot \frac{1}{\pi} \left\{ 2 \cos^{-1} \left(\frac{y}{2a} \right) - \left(\frac{y}{a} \right) \left[1 - \left(\frac{y}{2a} \right)^2 \right]^{1/2} \right\}$$

group 2 part = 1

n_1 = core RI

n = medium RI

y = lateral offset of core axes

a = fiber core radius

Lateral misalignment loss:

$$Loss_{lat} = -10 \log_{10} \eta_{lat} \text{ dB}$$

* Example 5.2: $n_1 = 1.5$, $n = 1$, $a = 50/2 = 25 \mu\text{m}$, $y = 5 \mu\text{m}$

Soln: (a)

$$\eta_{lat} = \frac{16(1.5/1)^2}{[1 + (1.5/1)]^4} \cdot \frac{1}{\pi} \left\{ 2 \cos^{-1} \left(\frac{5}{50} \right) - \left(\frac{5}{25} \right) \left[1 - \left(\frac{5}{50} \right)^2 \right]^{1/2} \right\}$$

$$= 0.203 \times (2 \times 0.9924 - 0.2 \times 0.9942) = 0.804$$

$$\text{Loss}_{\text{lat}} = -10 \log_{10} \eta_{\text{lat}} = -126.83 \text{ dB} \quad 0.944$$

(b) Index match means no air gap

$$\eta_{\text{lat}} = \frac{1}{\pi} \left\{ 2 \cos^{-1} \left(\frac{y}{2a} \right) - \left(\frac{y}{a} \right) \left[1 - \left(\frac{y}{2a} \right)^2 \right]^{\frac{1}{2}} \right\}$$

$$= \frac{1}{3.1416} \left\{ 2 \cos^{-1} \left(\frac{5}{50} \right) - \left(\frac{5}{25} \right) \left(1 - \left(\frac{5}{50} \right)^2 \right)^{\frac{1}{2}} \right\}$$

$$= 168.823 \text{ dB} \quad 0.873$$

$$\text{Loss}_{\text{lat}} = -10 \log_{10} \eta_{\text{lat}} = -122.26 \text{ dB} \quad 0.589$$

* Lateral Misalignment loss for multimode graded index:

Depends on RI gradient, $\alpha = 2$

$$\text{i) } L_T = \frac{2}{\pi} \frac{y}{a} \left(\frac{\alpha+2}{\alpha+1} \right) \text{ for } 0 \leq y \leq 0.2a$$

$$\text{ii) } \eta_{\text{lat}} = 1 - L_T$$

$$\text{iii) } \text{Loss}_{\text{lat}} = -10 \log_{10} \eta_{\text{lat}}$$

* with parabolic refractive index $\alpha = 2$,

$$L_T = \frac{2}{\pi} \frac{y}{a} \left(\frac{\alpha+2}{\alpha+1} \right)$$

$$= \frac{8}{3\pi} \frac{y}{a} = 0.85 \frac{y}{a}$$

* Modes for Lateral Misalignment:

i) Leaky mode; $L_T = 0.75 (y/a)$

ii) Guided mode; $L_T = 0.64 (y/a)$

iii) Leaky plus guided mode; $L_T = 0.5 (y/a)$

* Example 5.3: $\omega = 2$, $a = 25 \mu\text{m}$, $y = 3 \mu\text{m}$

Soln:

(a) For guided mode,

$$L_T = 0.64 \left(\frac{y}{a} \right) = 0.64 \times \frac{3}{25} = 0.0768$$

$$\eta_{\text{lat}} = 1 - L_T = 0.9232$$

$$\text{Loss}_{\text{lat}} = -10 \log_{10} \eta_{\text{lat}} = 0.467 \text{ dB}$$

(b) For both guided and leaky mode,

$$L_T = 0.75 \times \frac{y}{a} = 0.75 \times \frac{3}{25} = 0.09$$

$$\eta_{\text{lat}} = 1 - L_T = 0.91$$

$$\text{Loss}_{\text{lat}} = -10 \log_{10} \eta_{\text{lat}} = 0.409 \text{ dB}$$

ii) Angular misalignment:

Angular coupling efficiency,

$$NA = n_1(\Delta)^{\frac{1}{2}}$$

$$\eta_{ang} = \frac{16(n_1/n)^2}{[1 + (n_1/n)^4]} \cdot \left[1 - \frac{n\theta}{\pi n_1(\Delta)^{\frac{1}{2}}} \right]$$

where, θ = angular displacement in radians

Δ = relative RI difference

Smaller the value of Δ , larger insertion loss

$$Loss_{ang} = -10 \log_{10} \eta_{ang} \quad \eta_{ang} = \frac{16(n_1/n)^2}{[1 + (n_1/n)^4]} \cdot \left[1 - \frac{n\theta}{\pi n_1(\Delta)^{\frac{1}{2}}} \right]$$

* Example 5.4 : $n_1 = 1.48$, $n = 1$, $NA = 0.2$, $\theta = 5^\circ = 5\pi/180$

Solⁿ: For $NA = 0.2$, fiber;

$$\eta_{ang} = \frac{16(n_1/n)^2}{[1 + (n_1/n)^4]} \cdot \left[1 - \frac{n\theta}{\pi NA} \right]$$

$$= \frac{16 \times 1.48^2}{(2.48)^4} \cdot \left[1 - \frac{5\pi/180}{\pi \times 0.2} \right]$$

$$= 0.926 \times 0.861 = 0.797$$

$$Loss_{ang} = -10 \log_{10} \eta_{ang} = 0.983 \text{ dB}$$

For $NA = 0.4$ fiber,

$$\eta_{ang} = \frac{16 \times (1.48)^2}{(1 + 1.48)^4} \cdot \left[1 - \frac{5\pi/180}{\pi \times 0.4} \right]$$

$$= 0.926 \times 0.930 = 0.861$$

$$Loss_{ang} = -10 \log_{10} \eta_{ang} = 0.646 \text{ dB}$$

* Fiber Splices: A permanent joint formed between two individual optical fibers in the field is known as fiber splice.

- need connectors to join
- need melting element like electrical arc to melt the glass particles of the core.

* Splicing Types:

1. Fusion

2. Mechanical:

i) precision tube splice

ii) loose " "

iii) V groove " "

iv) precision pin " "

3. Multiple splices

* Fusion splicing: of single fiber involves the heating of the two prepared fiber ends to their fusing point with the electric arc between the two optical fibers.

* 2 types of machine

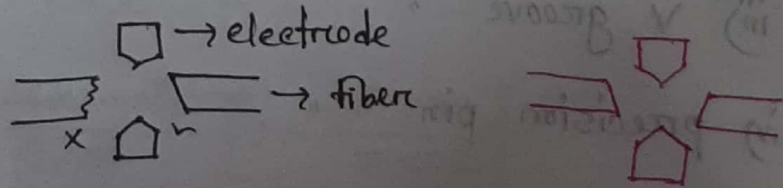
1. Table top model - display digital interface, control button, optical fiber

2. Handheld device - same "

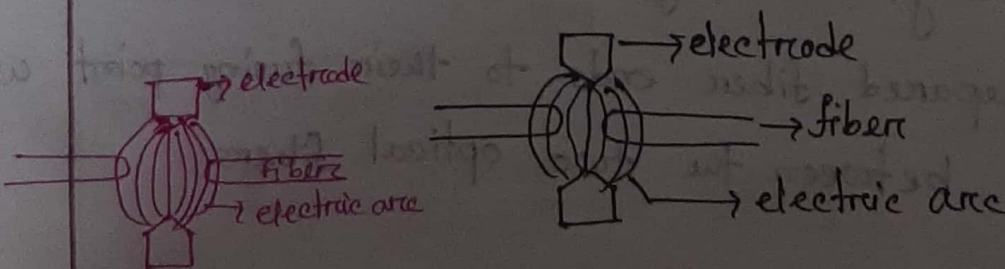
- monitor এ দেখা যায় core এর position ঠিক আছে কিনা
- fiber কে cutter machine দিয়ে কাটতে হয়। then yellow cover remove করে, machine দিয়ে cladding অংশে হয়। Then core join করা হয়
- monitor এ accuracy show করে

* Procedure:

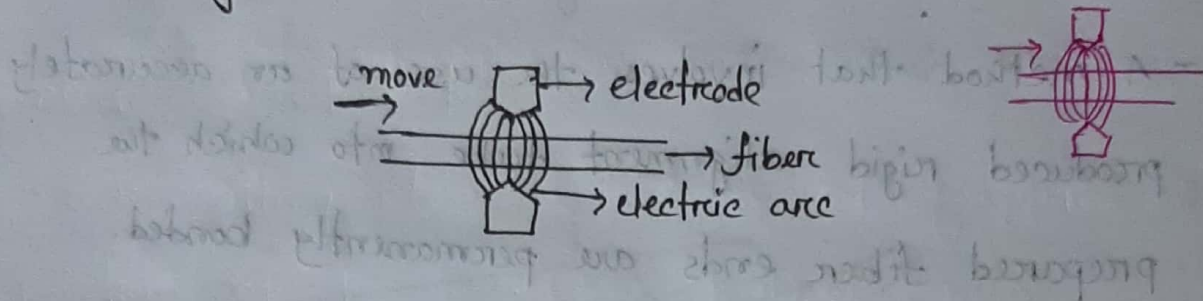
1. Initial setting: The fiber ends should be straight



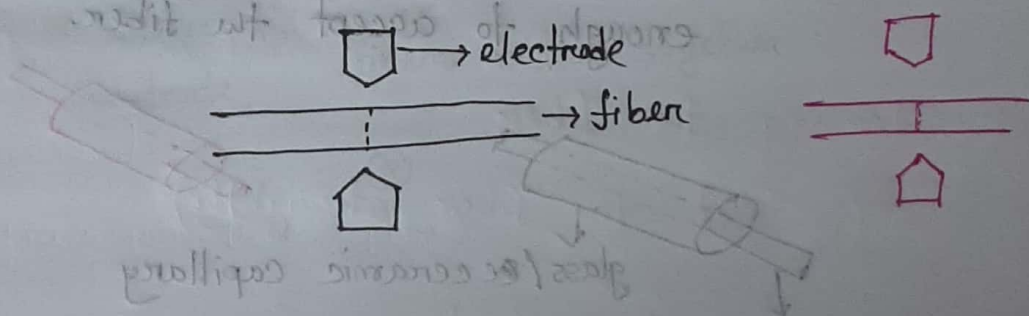
2. Arrangement of smooth surface by pre-fusion: will get the same heat on both sides.



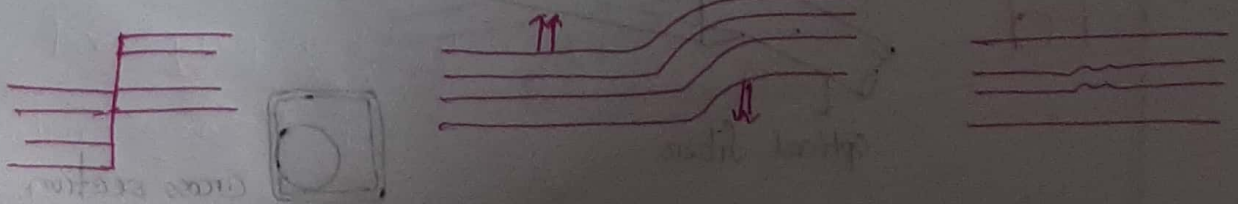
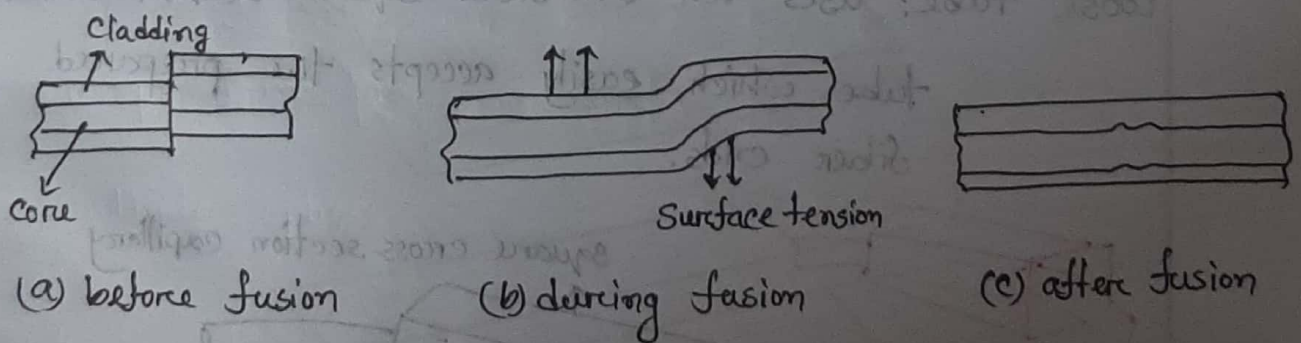
3. Pressed Together: The fibers are pressed together after observing the monitor.



4. Accomplishment of splice: A notification ^{arrives} comes when the fibers are joined after heating.



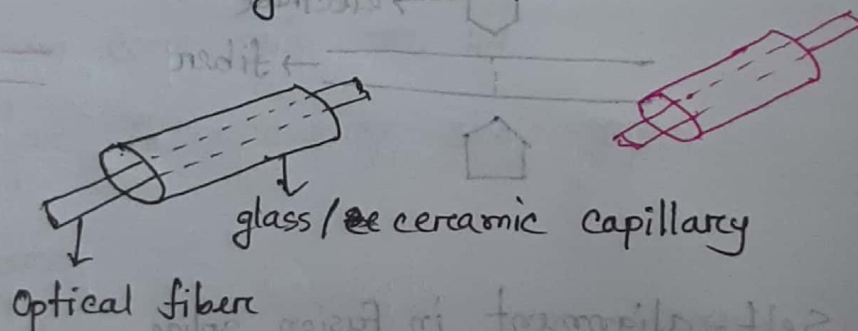
* mecha Self-alignment in fusion splice



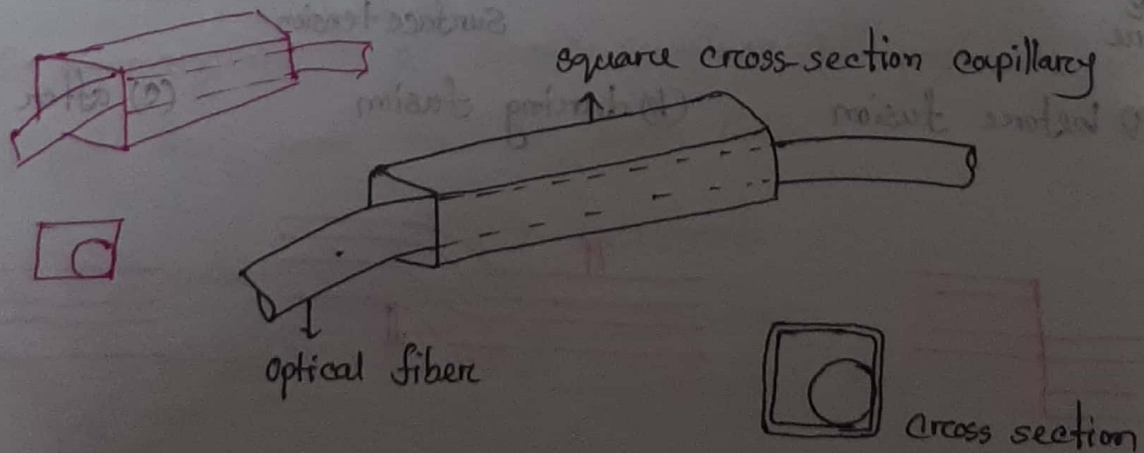
* Mechanical Splices:

- Tube splicing techniques: snug tube, loose tube
- A method that involves the use of an accurately produced rigid alignment tube into which the prepared fiber ends are permanently bonded.

Snug tube: utilize a glass or ceramic capillary with an inner diameter just large enough to accept the fiber.

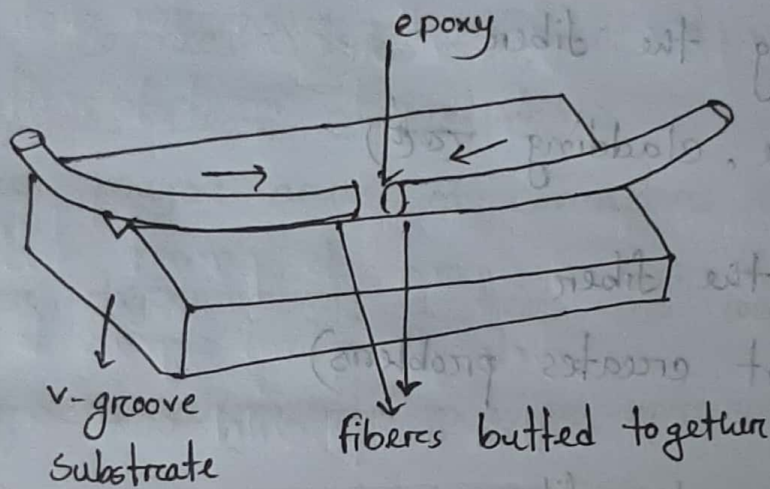


Loose tube: uses an oversized square section metal tube which easily accepts the prepared fiber ends.

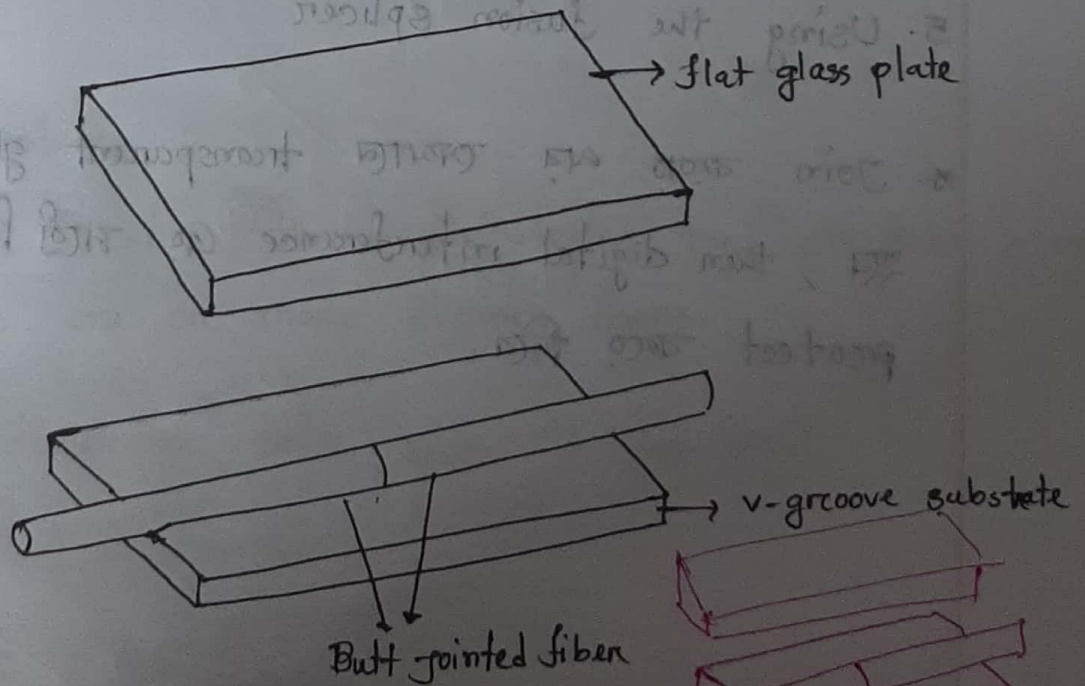


* V-groove splices

- Formed by sandwiching the butted fiber ends between a v-groove glass substrate and a flat glass retainer plate



(a)



(b)

*** Steps for joining

1. Adding the heat-shrink sleeve
(core cover and transparent part)

2. Stripping the fiber
(core, cladding coat)

3. Clean the fiber
(dust creates problems)

4. Cleaving the fiber
(to cut the fiber straightly)

5. Using the fusion splicer

* Join with a transparent glass for light
ray, then digital interference is made for core
protect with light.

* Fiber Connectors

- i) Cylindrical ferrule connector
- ii) Duplex & multiple fiber connector
- iii) Fiber connector types - ST, SC, LC, MU, MT-RJ, MPO/MTP
- iv) Expanded Beam connectors

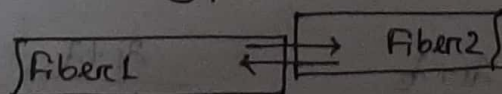
* Three major areas of connectors

1. Fiber termination: protects and locates the fiber ends
2. Fiber end alignment: provide optimum optical coupling
3. Outer shell: maintains the fiber connection and alignment, protects the fiber ends from the environment, provides strength to the joint.

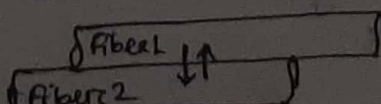
* Fiber Couplers

- A coupler is a device that distributes the light from a main fiber into one or more branch fibers.
- power transfer takes place in two ways:

i) Core interaction type



ii) Surface interaction type

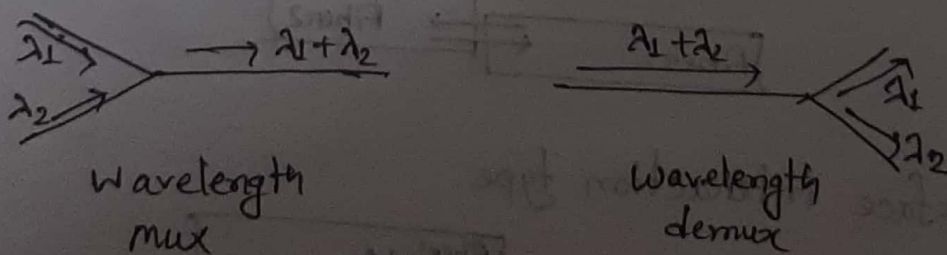
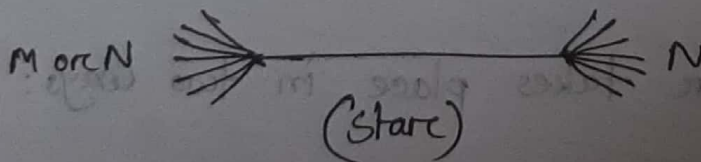
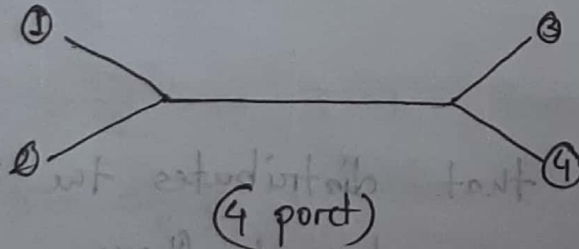
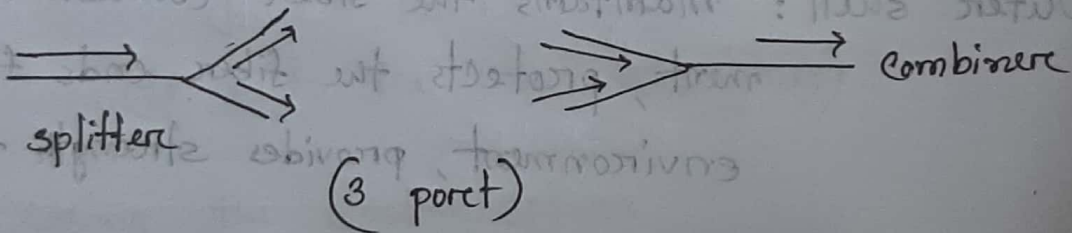


* Coupler types:

1. 3 and 4 port coupler: used for signal splitting, distribution and combining

2. Star coupler: generally used for distributing a single input signal to multiple outputs

3. WDM devices: designed to permit different wavelength optical signals to be transmitted in parallel on a single fiber.



* 3- and 4 port Couplers: 2 methods

1. Lateral offset method

2. Semitransparent mirror method

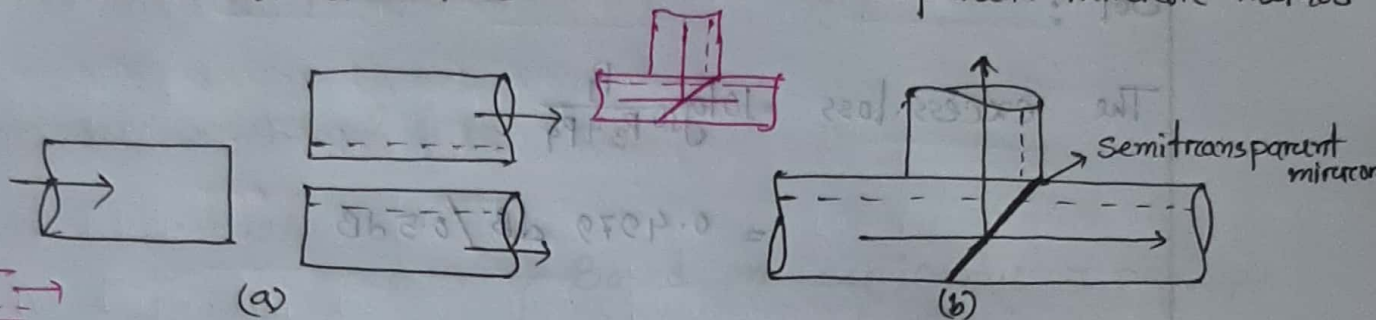


Fig: Fabrication technique for 3 port coupler

* Fused Biconical Taper Coupler (FBT)

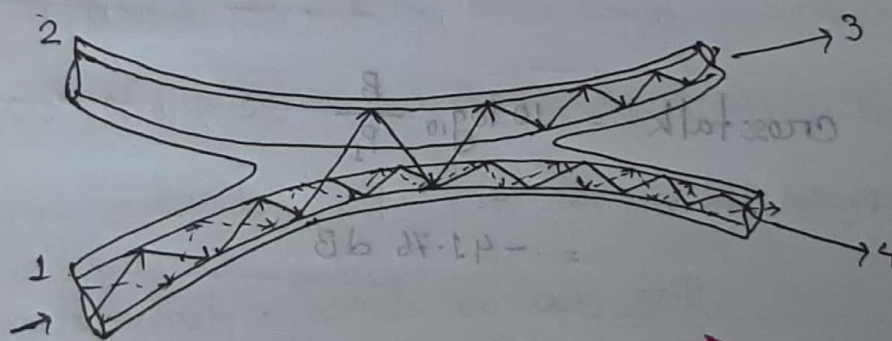


Fig: Structure of FBT

Excess loss (4 ports): $10 \log_{10} \frac{P_1}{P_3 + P_4} \text{ (dB)}$

Insertion loss (1 to 4) = $10 \log_{10} \frac{P_1}{P_4} \text{ (dB)}$

Crosstalk (4 port) = $10 \log_{10} \frac{P_2}{P_1} \text{ (dB)}$

Split ratio = $\frac{P_3}{P_3 + P_4} \times 100\%$

= $\left[1 - \frac{P_4}{P_3 + P_4} \right] \times 100\%$

$E.L = 10 \log \frac{P_1}{P_3 + P_4}$

$I.L = 10 \log \frac{P_1}{P_4}$

$C.T = 10 \log \frac{P_2}{P_1}$

$S.R = \frac{P_3}{P_3 + P_4} \times 100$

* Example 5.7 : $P_1 = 60 \mu W$, $P_2 = 0.004 \mu W$, $P_3 = 26.0 \mu W$, $P_4 = 27.5 \mu W$

Soln:

The excess loss = $10 \log_{10} \frac{P_1}{P_3 + P_4}$

$= 0.4979 \text{ dB} / 0.5 \text{ dB}$

The insertion loss = $10 \log_{10} \frac{P_1}{P_4} / 10 \log_{10} \frac{P_1}{P_3}$

$= 3.388 \text{ dB} / 3.63 \text{ dB}$

The crosstalk = $10 \log_{10} \frac{P_2}{P_1}$

$= -41.76 \text{ dB}$

The split ratio = $\left[\frac{P_3}{P_3 + P_4} \right] \times 100\%$

$= 48.59\%$

Chapter 06

Optical Sources

* Optical source 2 types:

1. Laser

2. LED

} Based on semiconductor principle

- electron passes/travels from one energy level to another;
this emits some photons which passes as the light.

- 2 process of this:

1. Radiative - emits photon in 2 possible ways

i) Spontaneously - emits on own will

ii) Stimulated - emits forcefully
Simulated

2. Non-radiative - creates heat

* Characteristics of Laser:

i) Highly coherent

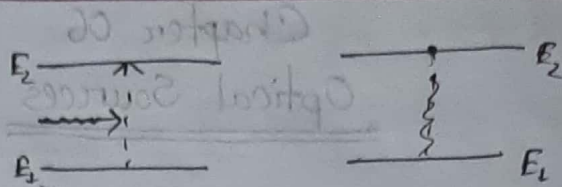
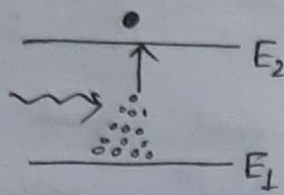
ii) Highly monochromatic

iii) Can be sharply focused

iv) Brightness - higher than any other light sources

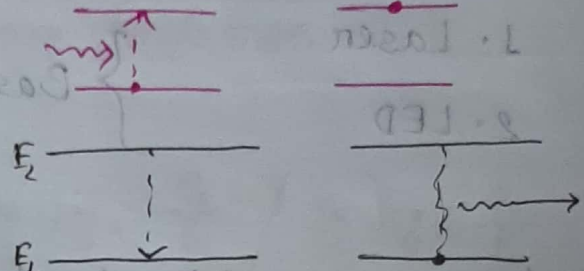
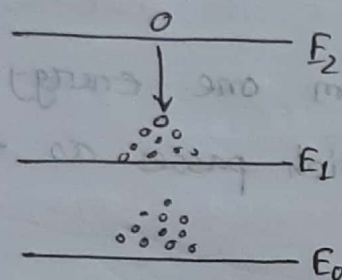
General Principles of Laser action:

a) Absorption:



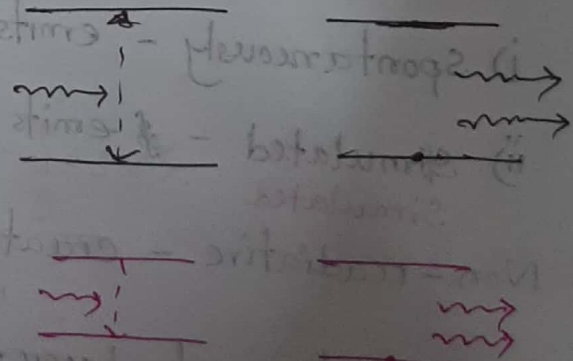
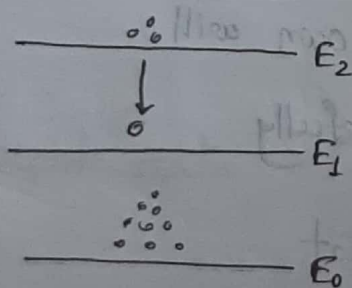
বাইরে থেকে energy absorb করে
higher level এ যাবে

b) Spontaneous Emission:

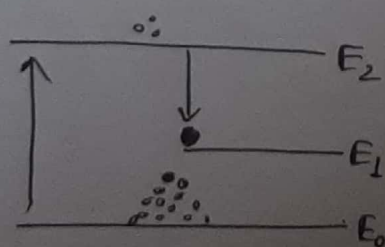


energy ছেড়ে দিয়ে নিচে আসবে

c) Stimulated emission:



d) Laser excitation:



* Absorption: The frequency of the absorbed or emitted radiation f is related to the difference in energy E between the higher energy state E_2 and lower energy state E_1 .

$$E = E_2 - E_1 = hf \quad h = 6.626 \times 10^{-34} \text{ J (Planks Const.)}$$

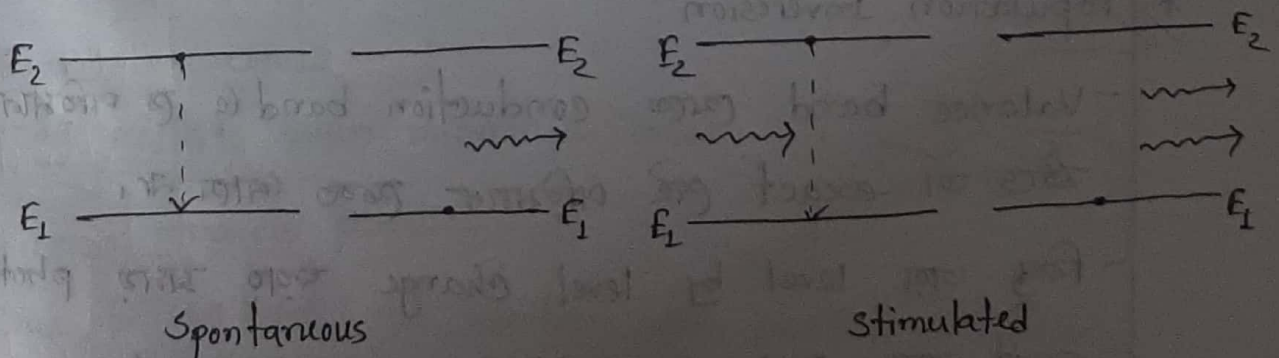
$$E = E_2 - E_1 = hf = h \times \frac{c}{\lambda}$$

(Energy absorb करे डीएर डीएर)

* Emission: Moves to the lower band by releasing the absorbed energy. 2 ways emission:

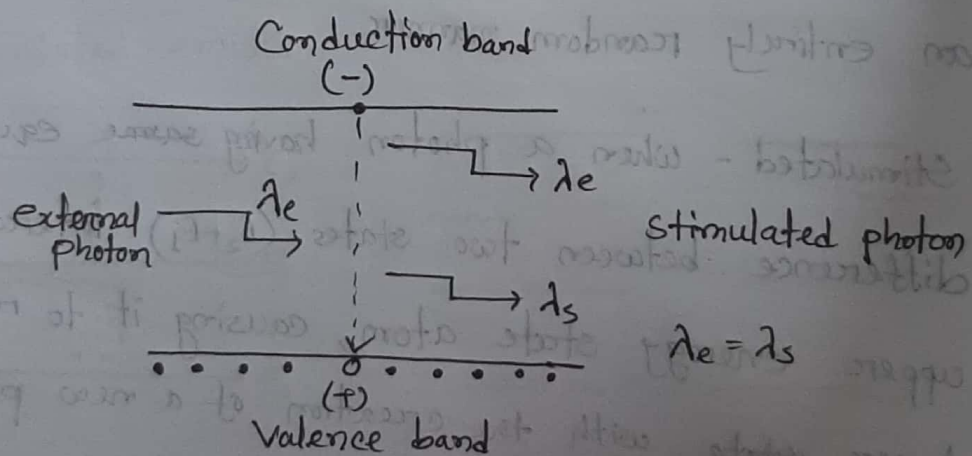
i) Spontaneous - atoms return to the lower energy state in an entirely random manner.

ii) Stimulated - when a photon having same equal energy to the difference between two states ($E_2 - E_1$) interacts with the upper energy state atom, causing it to return to the lower state with the creation of a new photon.



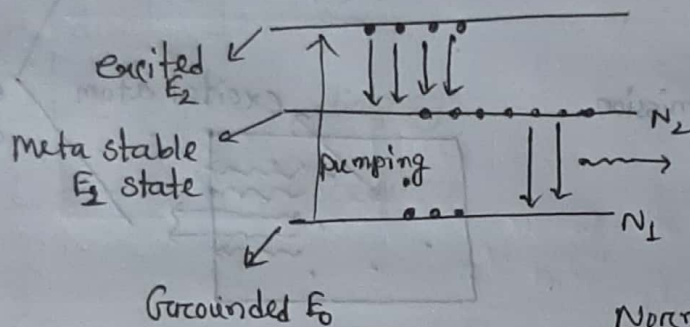
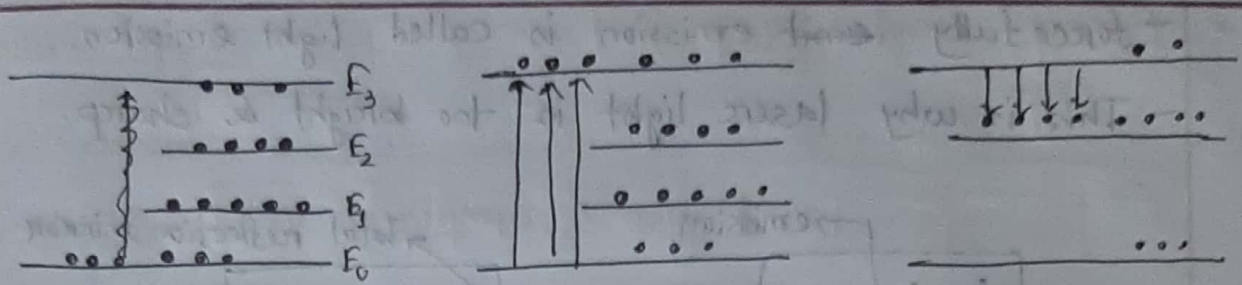
* Laser operation:

- 1) external photon hit করে মাঝে electron এর band এ আছে তা উপর depend করে band change করে।
Depending on where the external photon hits, electron will change its band.
- 2) 2 types bands; Valence (lower band), Conduction (upper)
- 3) Laser operation কে ভাব্য stimulate emission করে হয় কোন ২ টি photon পাওয়া যায়
- 4) বাইরে থেকে external photon বা force দিলে Conduction band থেকে valence band এ চলে আসবে।



* Population Inversion

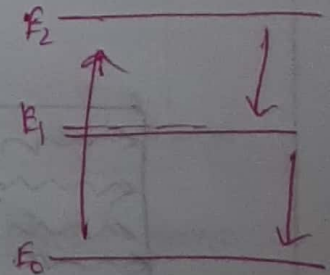
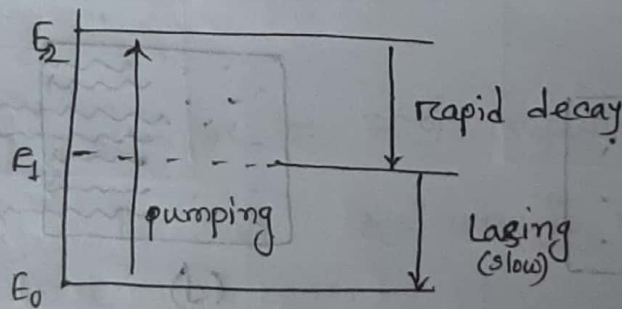
- Valence band থেকে conduction band এ যে পরিমাণ e উঠে তা exact সেই পরিমাণে ফেরত আসে না।
- কিন্তু তা level by level change করে সময় photon emit করে। But 2 Band কে e এর amount concept দিও থাকে না, inverse হয়ে যায়।



E_2 ৰে বহুদিন থাকে পাৰে বা

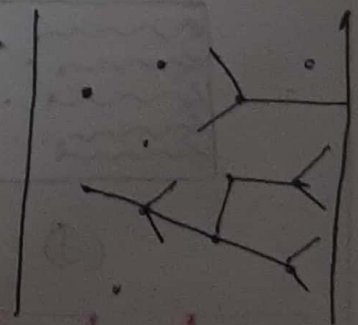
E_1 ৰা more time থাকে

Normal $N_1 > N_2$; Inverse $N_2 > N_1$

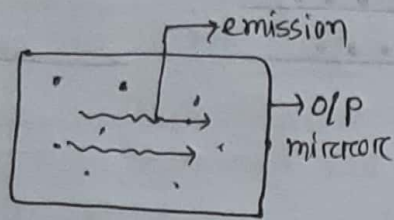


* Amplification of Photons:

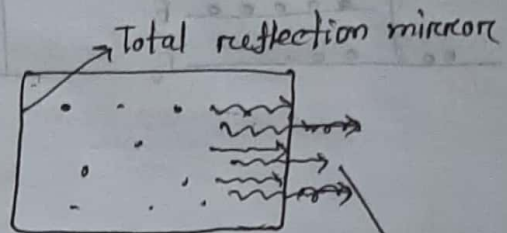
- When electrons are collided with e/o , 2 photon emits
- ray reflects when mirror is used
- Photon travels parallelly
- Due to collision, the electrons gets excited to reflect
- energy increases with increasing photons
- When there's many photons, precise point can be get. This points emits forcefully from this point.



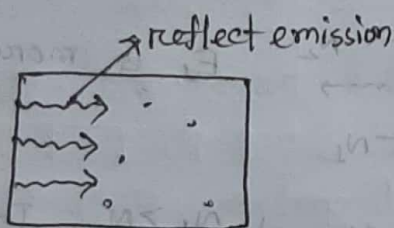
- forcefully emit emission is called light emission.
- This is why laserc light is too bright & sharp.



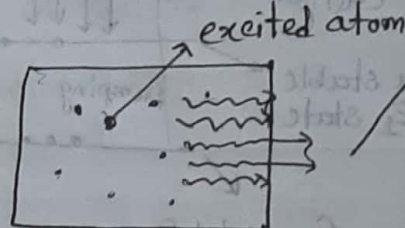
(a)



(b)

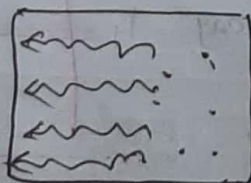


(c)

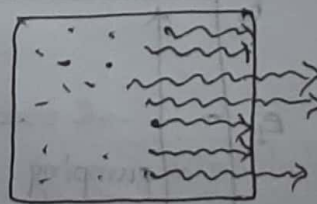


(d)

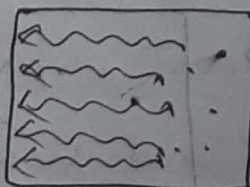
Light
escaping
cavity



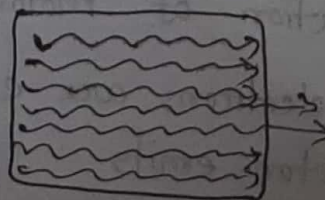
(e)



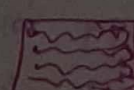
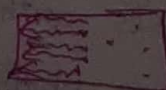
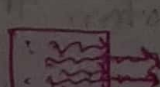
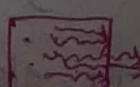
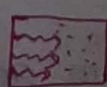
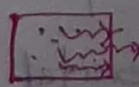
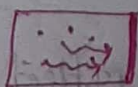
(f)



(g)

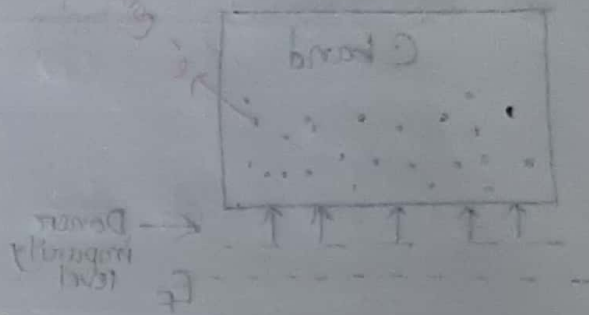


(h)

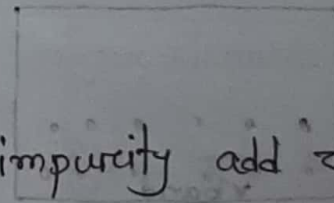
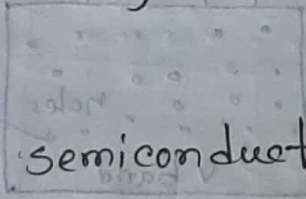


* Laser application:

1. Medicine
2. Industry
3. Defense
4. Research
5. Product development
6. Laser Lighting displays
7. Laser skin procedure



* LED vs LD
(spontaneous) (stimulated)



* LED - Semiconductor $\Rightarrow$ impurity add করে

- electron & hole একে আকর্ষণে attract করে
- n-type p-type create করে

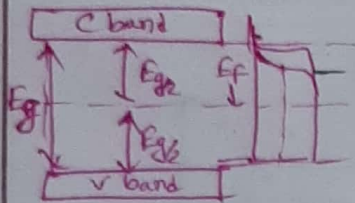
n-p junction $\Rightarrow$ forward/reverse bias করা হয়।

Forward $\rightarrow$ P-P, n-n

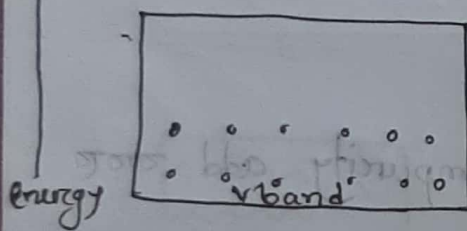
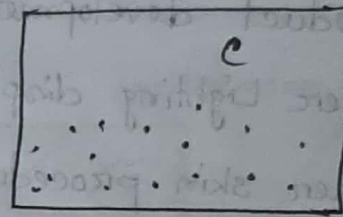
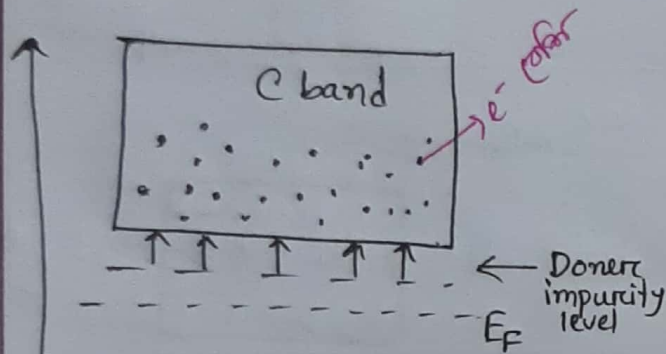
Backward $\rightarrow$ P-n, n-P



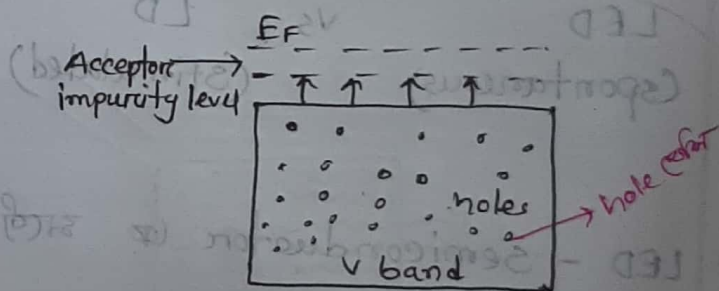
* Energy band: gap between valence band and conduction band (E_g)



band gap & impurity level
Donor impurity, acceptor impurity



n-type

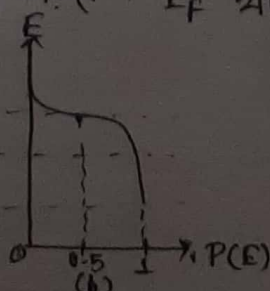
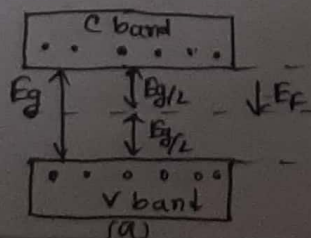


p-type

E_f = Energy Fermi level

- impurity level add করে add করে হয় যে donor impurity level থেকে C band এ যাবে; e^- বাড়বে। E_f থাকবে C band এর দিকে

- acceptor impurity level add করলে V band এর e^- C band এ যাবে, hole বাড়বে। E_f থাকবে V band এর দিকে



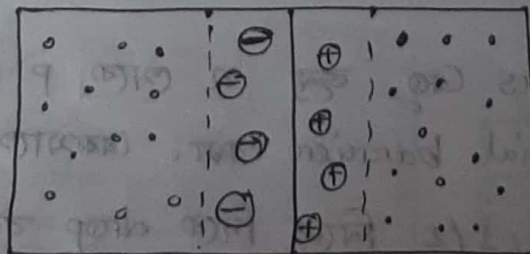
- * Fermi level সর্বদা সমান
- * equal no of hole & e^-
- * (b) is prob. distrib. corresponding to (a)
- * prob. drop করে থাকে

* P-n junction

- p, n type diode एक व्यवस्थित चारा
- Conductor দিয়ে junction ছেঁই কল current pass হলে heat transfer হয়
- semi-conductor হলে photon generate হয়, যা laser হিসাবে দেখা যায়।

* Depletion Layer

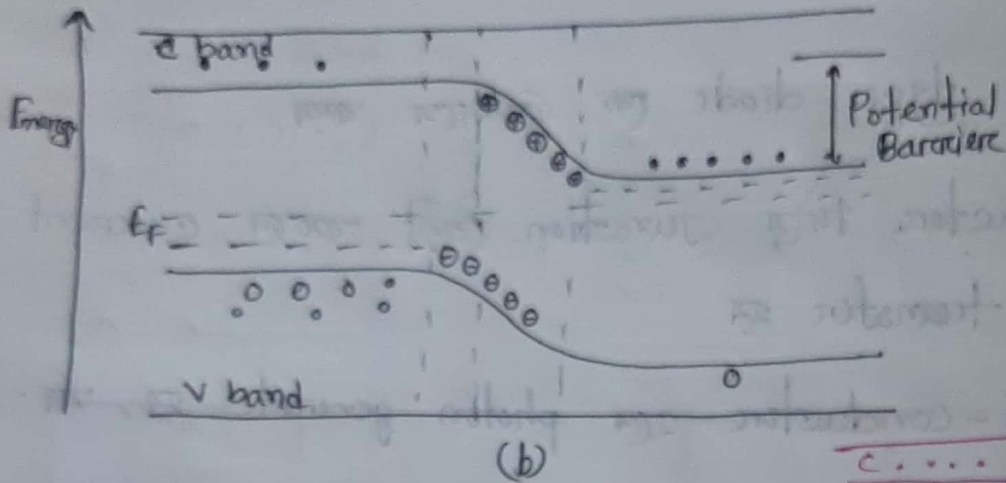
- middle layer between p-type and n-type
- p type major-holes ; n type major electrons
- forward bias p-p, n-n; layer কমে যায়, e^- and h movement করতে পারে, movement এর সময় recombination হয়। e^- h মিলিত হওয়ায় সময় photon release হয়, যা light হিসাবে পাওয়া যায়।



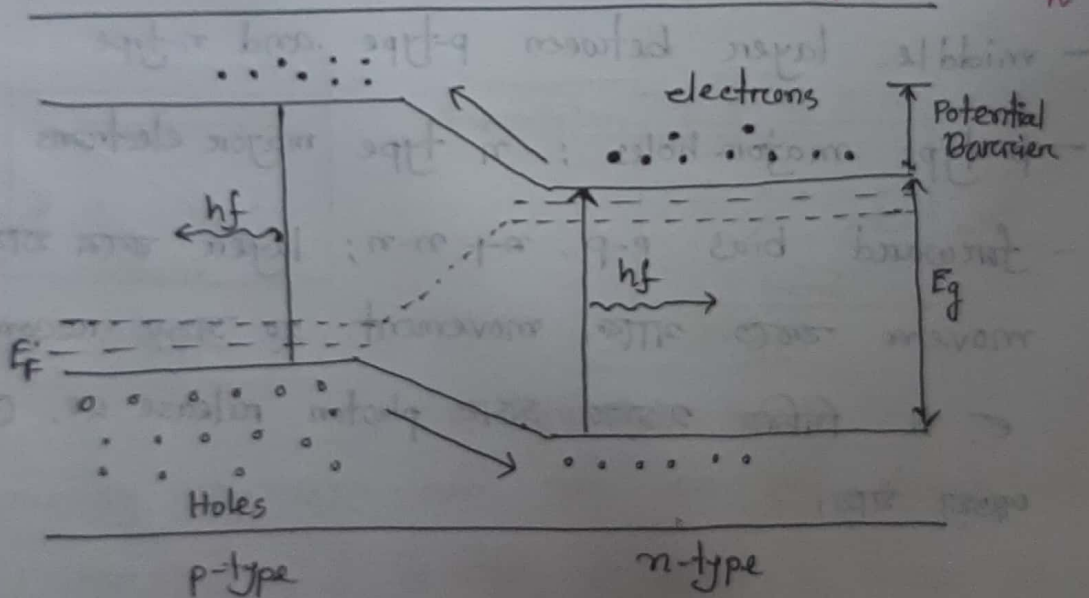
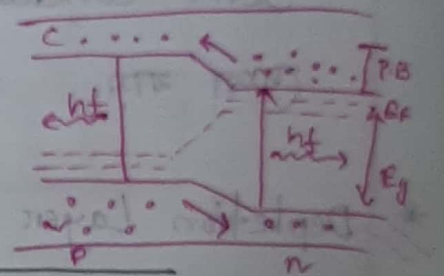
p type

n type

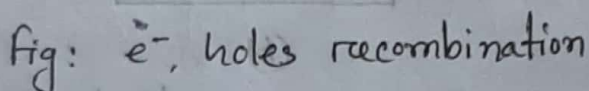
(a)



* p-n junction with forward bias



- forward bias তে n থেকে p তে electron যাবে কারণ potential barrier কম, একসাথে অনেকগুলো e^- যাবে.
- যাবার সময় $1/2$ নিজে দিকে ঝেঁপে যাবে.
- Holes গুলো p থেকে n এ যাবার ক্ষমতা কিছু উল্লেখ আছে.
- e^- & holes recombine হয়ে photon create করে যা light হিসেবে দেখা যায়.

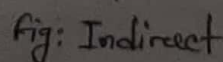
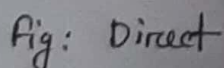


1) Radiative : band gap ko equal energy वि. photon emit kr

Band energy, $E_g = E_2 - E_1 = hf = \frac{hc}{\lambda}$

2 types transition -

- i) Direct Bandgap semiconductor
- ii) Indirect

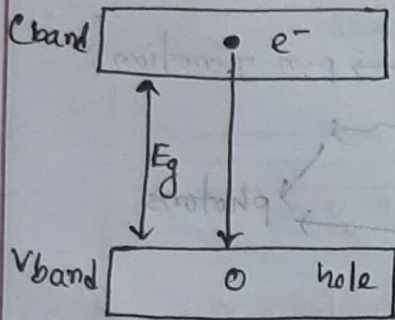


- momentum or wave vector (k) change Δk
- e^- directly drop Δk , slightly displace Δk
- slow

Ans. difference श्य. यदि C band & V band Cao

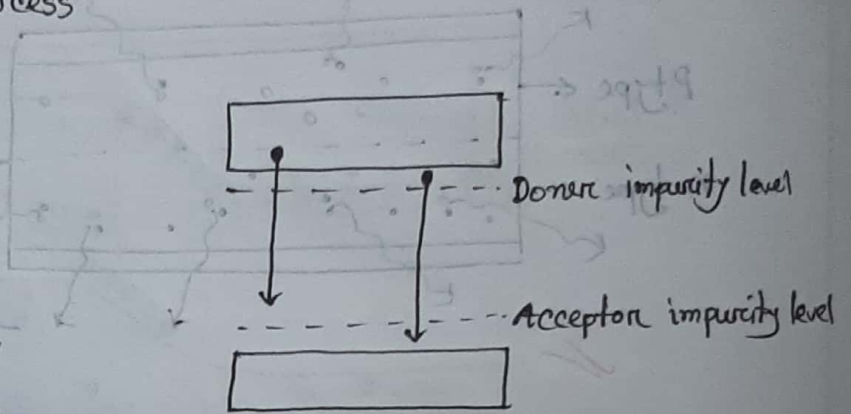
Const same 2x constant 1 Co learner & 1 Co learner constant error & Drop directly

* Recombination Process



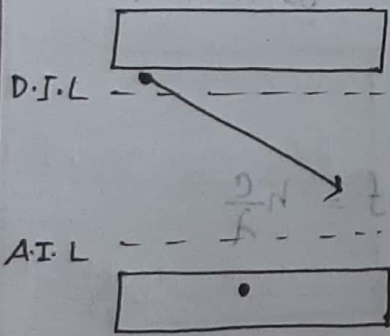
(a)

band-to-band transition



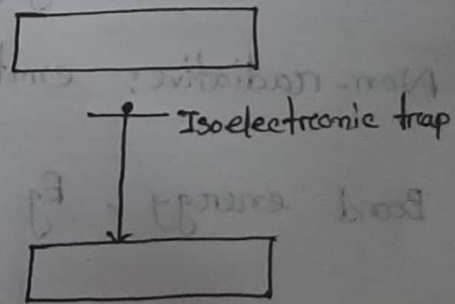
(b)

cband to acceptor impurity & donor impurity to vband



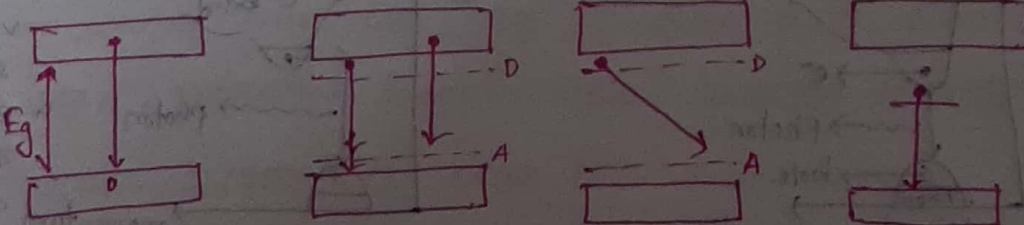
(c)

Donor impurity to acceptor impurity transition



(d)

recombination from an Isoelectronic impurity to v band



* Stimulated Emission & Lasing

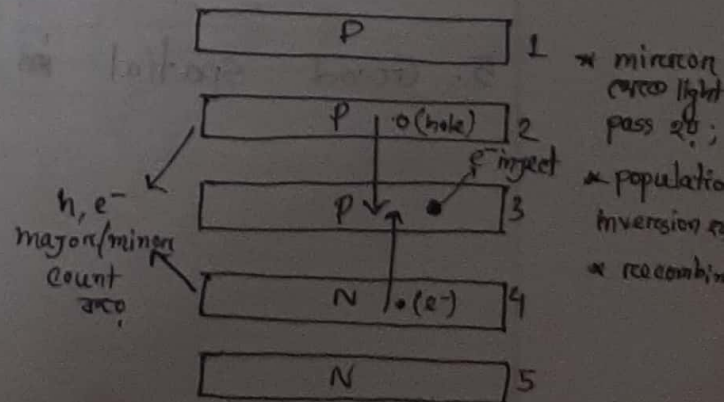
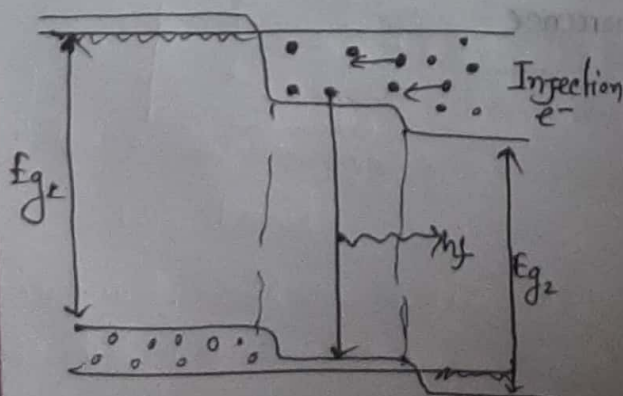
- uses population inversion
- external α photons goes to c band
- But when c band drops, 2 or 3 layer gets that
- layer drop may be rapid or slow
- Drops with more force if the distance is larger
- e^- goes to upper layer due to applying force and emits light when drops to lower layer. This emission of light is called stimulated emission.

* Hetero junction

- not only p-n junction is there, There is 3 or 5 type of p-n junction.

p p n
↓
cleaved
mirror

p p p n n
↓
cleaved
mirror

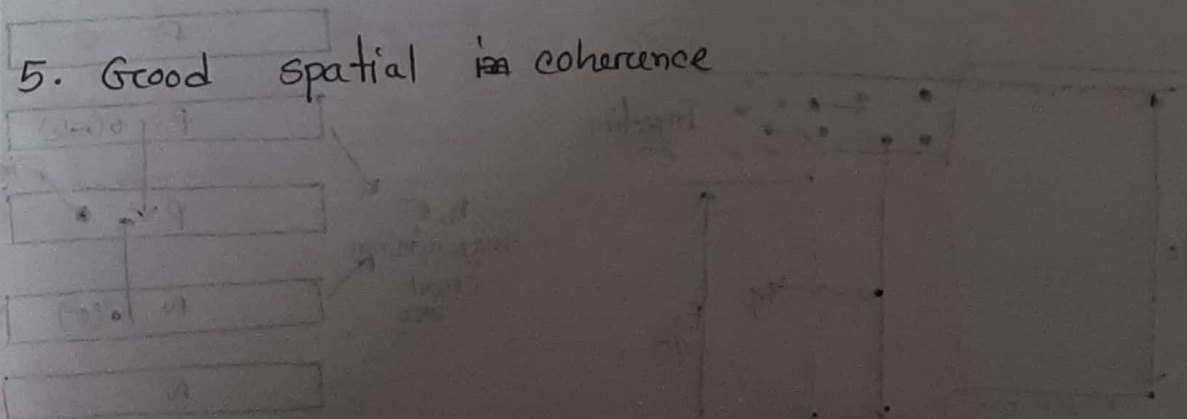


* Semiconductor materials criteria:

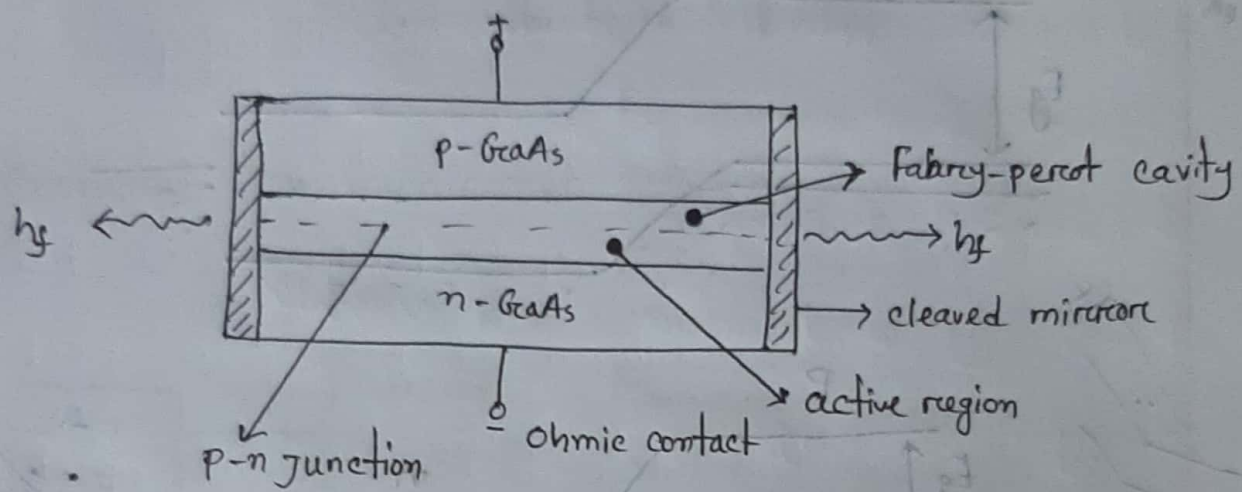
1. P-n junction formation - formation with suitable characteristics for carrier injection
2. Efficient electroluminescence - devices fabricate should have ^{high} probability of radiative transitions and a high internal quantum efficiency. Either direct or indirect bandgap semiconductors
3. Useful emission wavelength - must emit light at a suitable wavelength to be utilized with current optical fiber and detectors.

* Advantages of semiconductor sources:

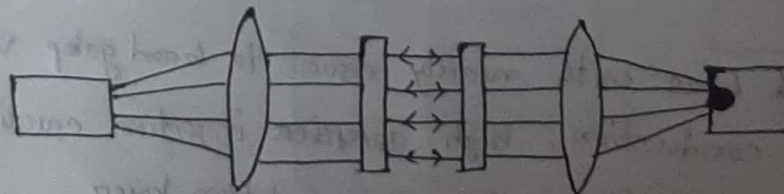
1. High radiance
2. Narrow line width
3. Modulation capabilities
4. Relative temporal coherence
5. Good spatial coherence



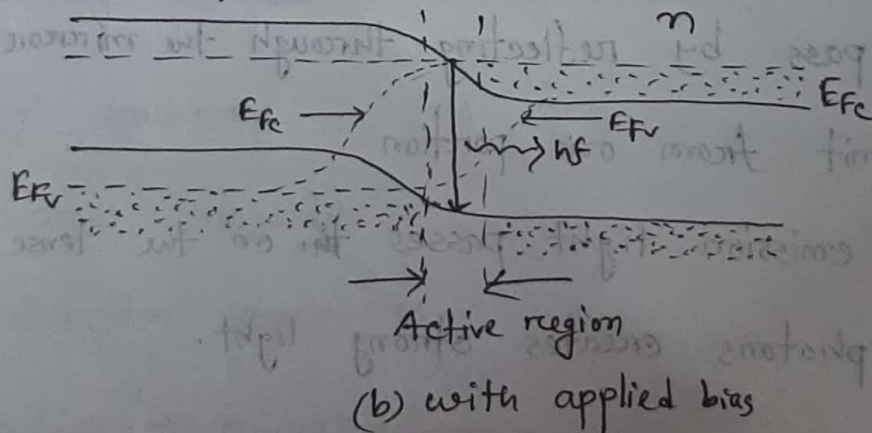
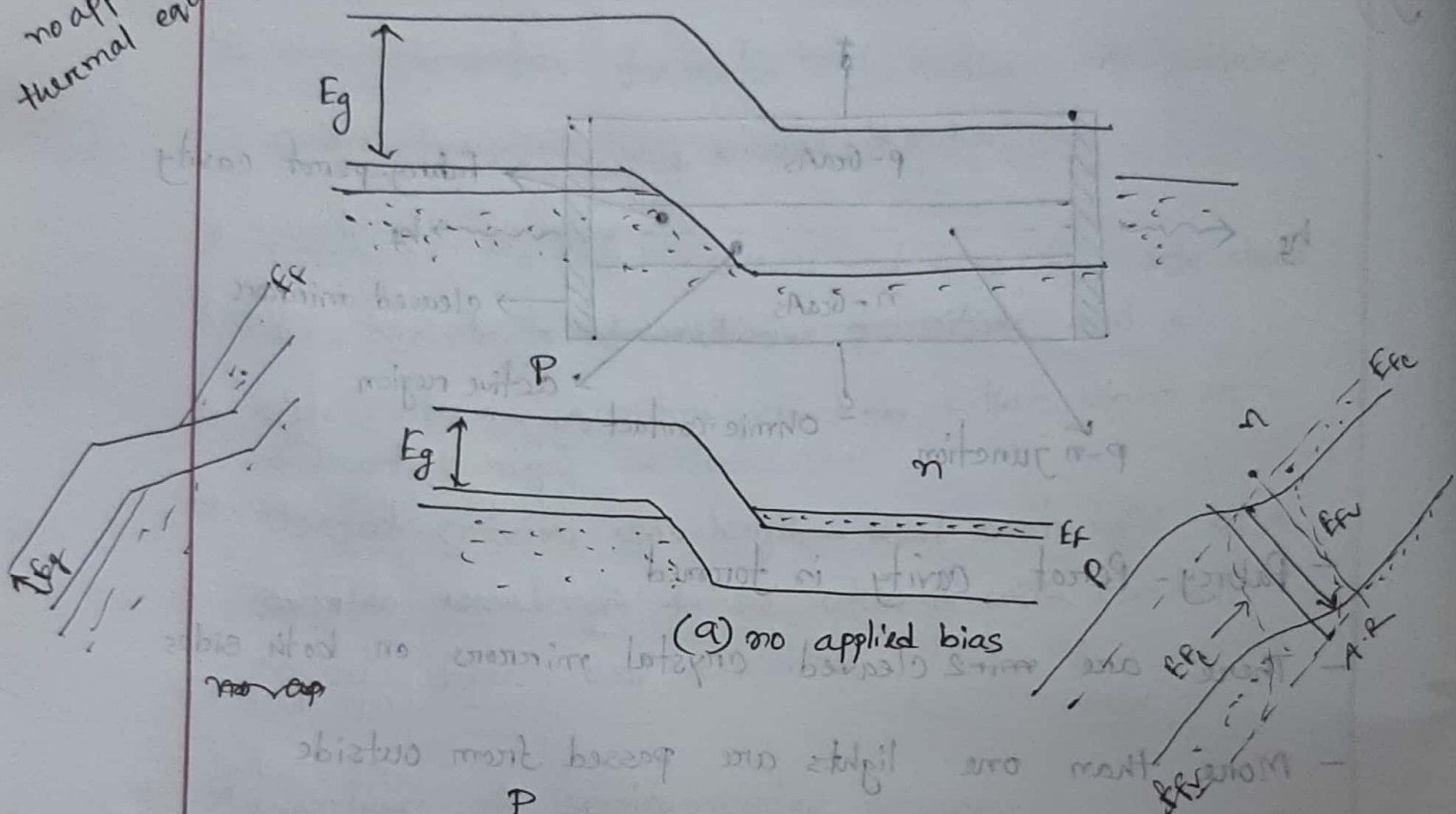
* Homo Junction



- Fabry - Perot cavity is formed
- There are ~~mir~~ 2 cleaved crystal mirrors on both sides
- More than one lights are passed from outside
- Lights pass by reflecting through the mirror
- Will emit from one portion
- after emission light passes ~~th~~ on the lense in front.
- more photons creates strong light.



no applied bias, thermal equilibrium, E_F is always same throughout the material



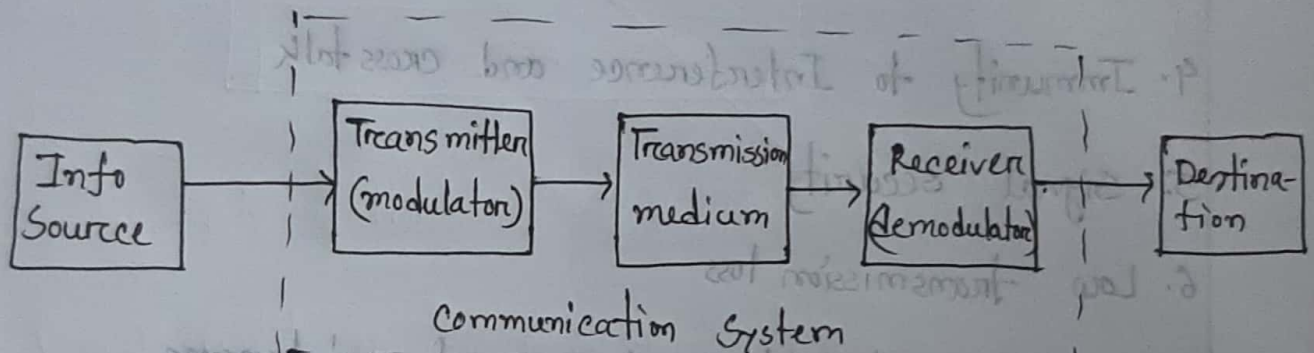
- forward bias with nearly equal to band gap voltage is applied
- direct conduction; high carrier injection carrier density
- there's an active region near depletion layer
- depletion layer contains simultaneously degenerate population of e^- & holes
- This satisfies stimulated emission

$$E_g/h < f < (E_{Fn} - E_{Fp})/h$$

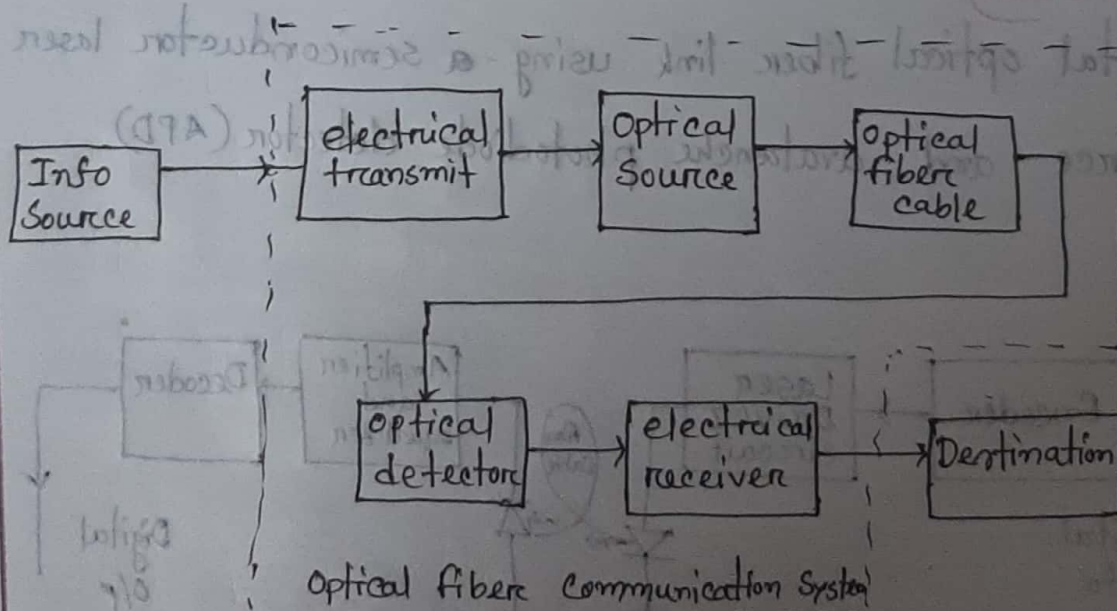
Chapter 01

Introduction

* General communication system:



* Optical fiber Communication System

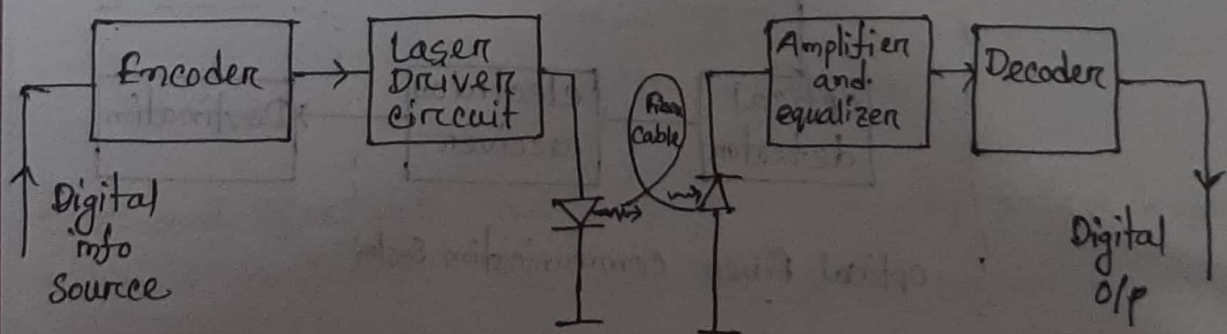


electrical signal $\rightarrow$ elec. stage (modulation) $\rightarrow$ LD/LED $\rightarrow$ transmission medium
 $\downarrow$
Destination $\leftarrow$ electrical stage (demodulation) $\leftarrow$ receiver

* Advantages:

1. Enormous potential bandwidth
2. Small size and weight
3. Electrical Isolation
4. Immunity to Interference and cross talk
5. Signal security
6. Low transmission loss
7. System reliability and ease of maintenance
8. Potential low cost
9. Flexibility

* Digital optical fiber link using a semiconductor laser source and avalanche photodiode detector (APD)



Optical Fiber Communication

* Why

- data faster transfer
- bulk data transfer
- long distance transmission/communication

Industry implication

- ISP also provide DSP (Data Service Provider)

KYC → Know Your Customers data

communication media হিসাবে optical fiber use করা হয়
central server থেকে

TCP/IP to optical fiber converter

- 1. Link3
 - 2. Banglaphone
- } DSP/ISP

Satellite parameters - 2

- 1) Uplink
- 2) Downlink

- data transfer transfer

- bulk data transfer

- long distance transmission / communication

Interconnected implication

- ISP also provide DSP (Data service Provider)

KYC → Know Your Customer data

Communication media fiber optic fiber use and so

Central server also

ISP/DP to optical fiber converter

1 link
2 link } ISP/DP

Subsea Optical Network

- responsible for transmission of 90% data
- for long distance transmission
- 366 subsea cables; 1006 cable landing stations
- proposed for 30 more
- needs repeater as a booster

Benefits:

- i) more durable
- ii) much cheaper
- iii) more info is carried
- iv) more secured in reliability

*How do we communicate with submarines?

⇒ It is difficult because radio waves do not travel well through good electrical conductors like salt water. The solution is to surface and raise an antenna above the sea level, then use ordinary radio transmissions. However, a submarine is vulnerable when on the surface. So by using optical fibre network, we can communicate with submarines.

* 2 টি PC-র মধ্যে connectivity ঠিক আছে কিনা সেটা monitor করতে হবে -

1. Cmd to ping করে।

2. Traceroute (trace route) दिल total route दिल एकर दिल
लवकर या कम route वा hop - 4 problem.

* Network Monitoring tools -

- 1. Solarc wind
- 2. Nagios

} with ping concept for checking

* কোন submarine cable ৬ problem রয়েছে তা হলো যাঃ

Network management center - ৩ submarine cables, repeaters
এবং কিছু information থাকবে।
কোন cable damage হলে বিনা ছাফর হয় বলা possible. Cable ৬
হোক problem হলে signal চিহ্নিত pass করতে না, আর তা কোন
repeater থেকে pass হচ্ছে না মনে network management center
থেকে জানা যায়।

Deep sea to cable unarmoured था,

Surface ko rakshak archmoured cable hai.

* Upgrading Capacity on Existing Submarine networks:

- a very cost effective means of adding high speed capacity

- increase the channel bit rate on an existing lit fiber pair

- increase the channel count on an existing lit fiber pair

- slot provisioning

- with piggybacking for checking

Optical Fiber Communication

* Network Terminology

Stations - devices that network subscribers use to communicate

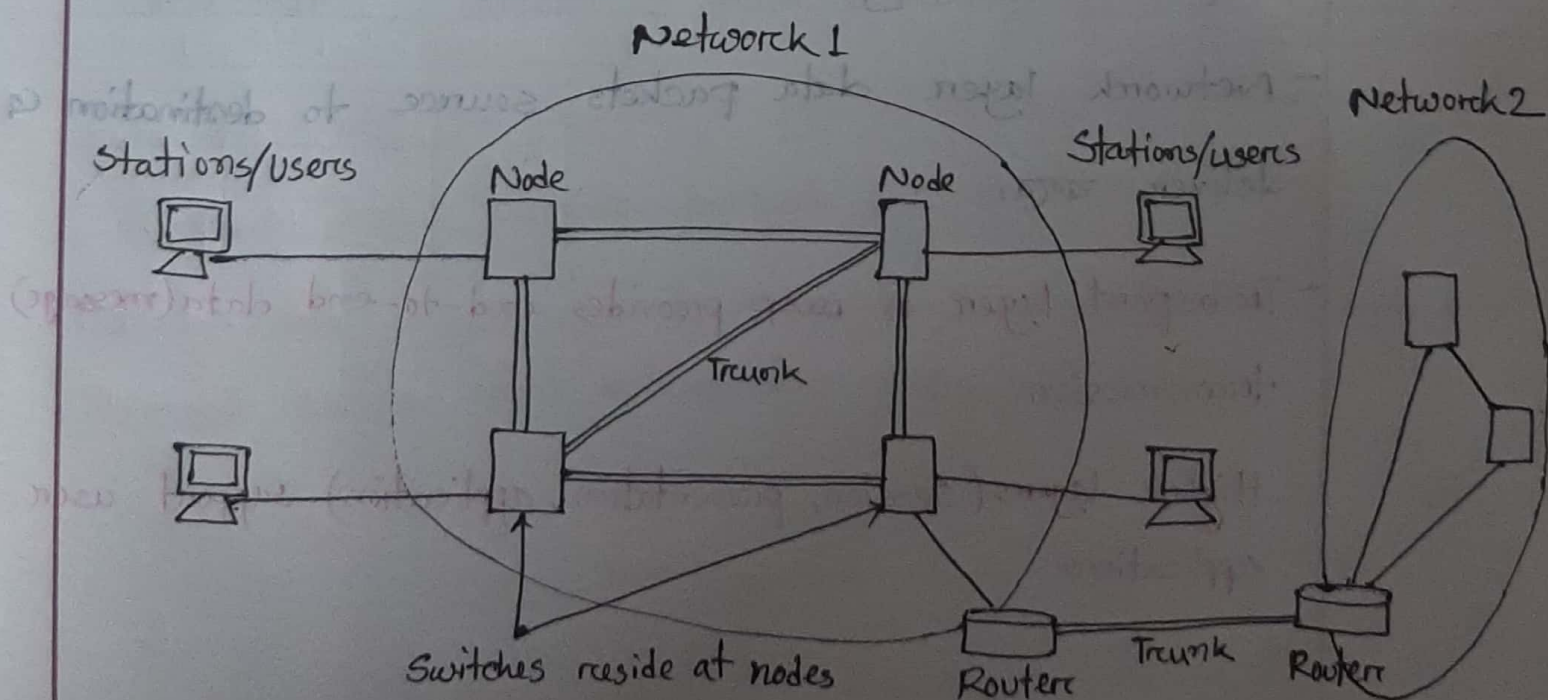
Ex - PCs

Network - collection of interconnected stations

Node - a point where one or more communication lines terminate

Trunk - transmission line that supports large traffic loads.

Topology - the logical manner in which nodes are linked together by information transmission transmitting channels to form a network.



* Segments of a Public Network

- local area network (LAN)
- campus network (extension of LAN)
- metro network (MAN)
- access network
- Wide Area Network (WAN)

* Protocol Stack Model

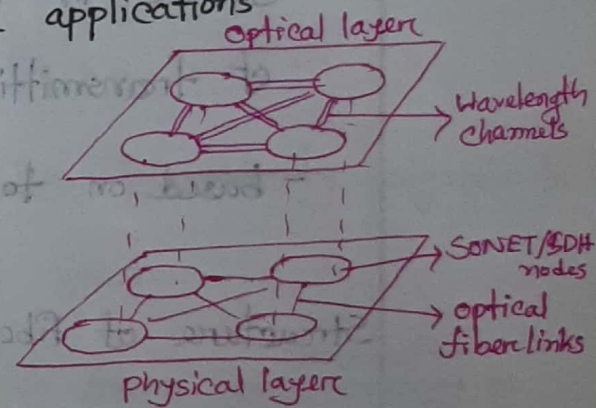
- Fiber optics physical layer & वायु वर्ण; वायुन - bit wise transmission media & आवृत्ति data pass करे
- Data link layer & link establish, maintain and release करे या directly 2 or node को connect करे.
- Network layer data packets source to destination & deliver करे;
- Transport layer ~~to~~ provides end-to-end data (message) transmission
- Higher layers (session, presentation, application) support user applications.

* Network Layering Concepts

- Network architecture: general physical arrangement
 - Protocols: A set of rules and conventions that governs the generation, formatting, control, exchange, and interpretation of ~~family~~ information that is transmitted through a telecommunication network or that is stored in a database.
 - Protocol stack: subdivide protocol into no. of individual layers
- Lower layer governs communication facilities.
Upper layer support user applications

* Optical Layer (used to describe network functions or services)

- wavelength based concept
- Just above the physical layer
- processes include wavelength multiplexing, adding and dropping wavelengths, support of optical switching. (networks that have these optical layer functions, are called wavelength-routed network)
- The physical layer provides a physical connection between two nodes, the optical layer provides lightpath services over that link.
- A lightpath is an end-to-end optical connection that may pass through one or more intermediate nodes.

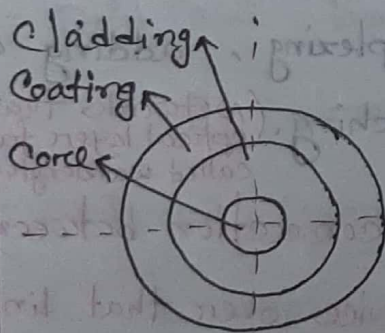


Introduction to Optical Fiber Optics

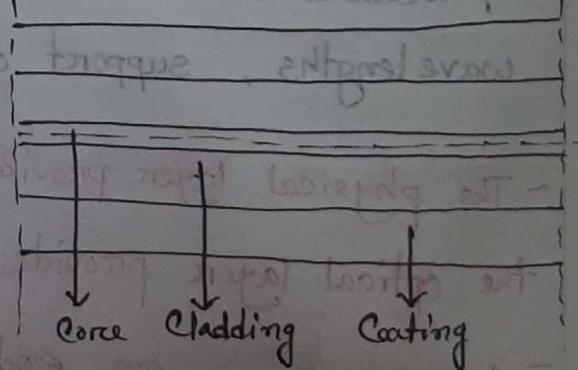
Fiber optic communication

- trapping light inside an optical fiber
- can carry any form of information
- fiber is an optical medium, means it is capable of transmitting light
- based on total internal reflection (TIR)

Structure of fiber optic cables



Transverse view



Longitudinal view

*** How lights travel through fiber

** Refraction, Refractive index

** TIR

* Types of Fiber-optic Cables

- i) Single-mode fiber
- ii) Multi-mode fiber
- iii) Step index fiber
- iv) Graded index fiber

** Fiber-optic Communication System

Transducer - a device which converts electricity signals to the corresponding light signal.

- a device that converts from one ^{form of} energy to another.

- Devices similar to light bulbs are used as optical transducers at the input of fiber-optic cables to convert electricity into light

- * Light Emitting Diode (LED)
 - * Laser Diodes (LD)
- } Fiber optic input
सिग्नल use करता है

** Wavelength Division Multiplexing

* Fiber-optic Technology

Benefits:

1. High transmission rate
2. Immunity to high electromagnetic interference
3. Low attenuation
4. High security
5. Small weight and size
6. Low power consumption

Drawbacks:

1. High installation cost
2. Difficulty in splicing

* Commercial Cables

- Medicine (Endoscopy)
- Telecommunications
- Computing
- Computer networking
- Transoceanic cables

Syllabus

Chapter 04 : Optical Fibers

Chapter 05 : Optical Fiber Cables

Chapter 06 : Light Sources and Transmitters

Chapter 07 : Photodiodes and Receivers

Chapter 08 : Connectors and Splices

Chapter 09 : Passive Optical Components

Chapter 10 : Active Optical Components

Chapter 11 : Optical Amplifiers

Chapter 12 : Wavelength Division Multiplexing

Chapter 05 - 12 ; 20 (Final)

Mid term → 05-08

Introduction to Optical Fibers

* **Fiber-optic Communication**: is a method of transmitting information from one place to another by sending pulses of infrared or visible light through an optical fiber. The light is a form of carrier wave that is modulated to carry information.

* It works based on Total Internal reflection (TIR).

* What is **Total Internal Reflection**?

Ans: TIR may be considered as extreme case of refraction. When a light ray strikes a boundary of two materials with different refractive index, it bends or refracts to an extent that depends on the ratio of the refraction index of the two materials.

* What is **Refraction**?

Ans: Refraction is the change in the direction of light passing from one medium to another.

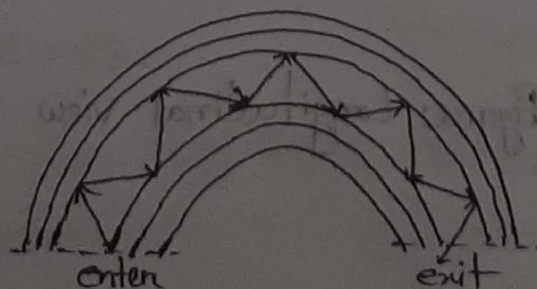
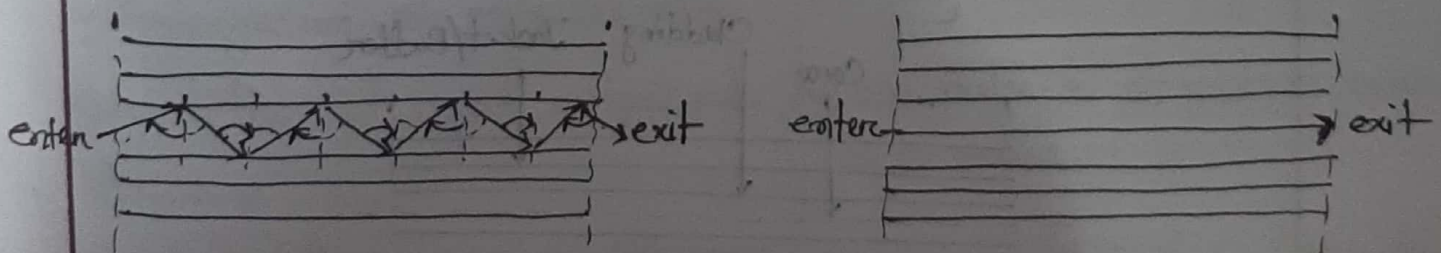
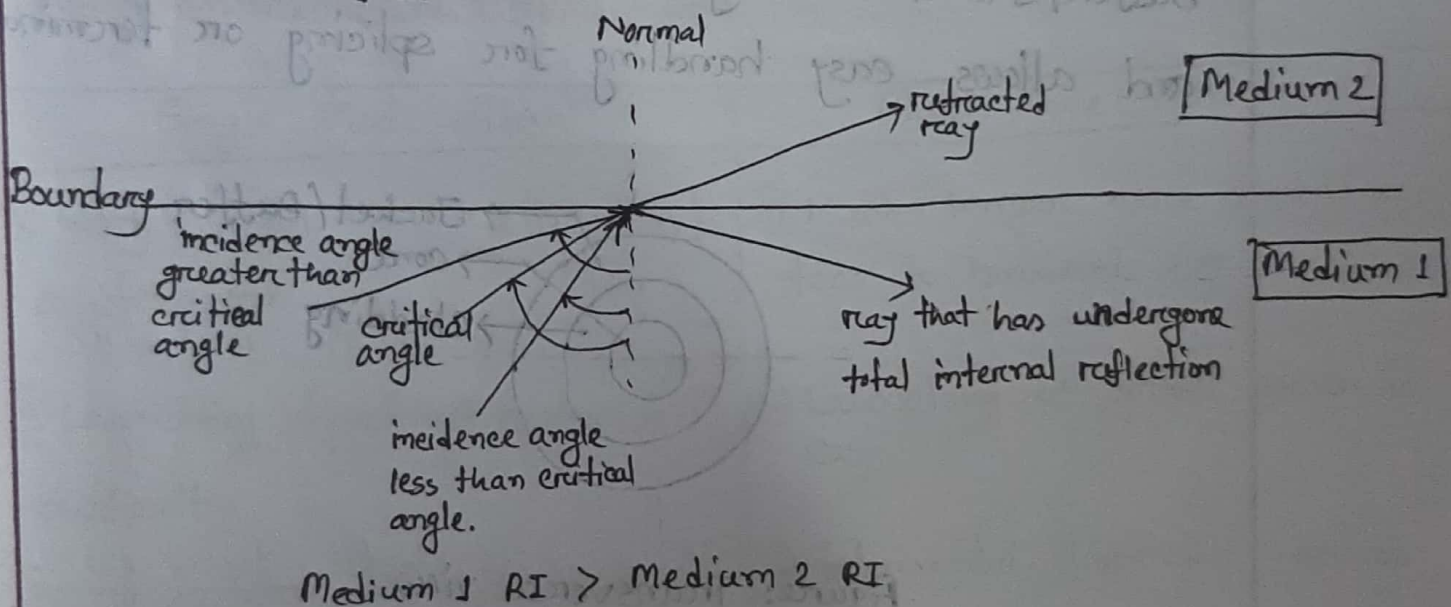
* What is **Refractive index**?

Ans: The refraction index of an optical medium is a dimensionless no. that gives the indication of the light bending ability of that medium. It determines how much the path of light is bent or refracted when entering a material.

**** Refractive index of an optical medium = speed of light in a vacuum (3×10^8 m/s) / speed of light in the optical medium**

Example- calculate the RI of an optical medium if light travels at a speed of 2.4×10^8 m/s inside the medium.

Solⁿ : $RI = 3 \times 10^8 / 2.4 \times 10^8$
 $= 1.25$



* **Optical fiber:** uses the principle of TIR to capture the light that is being transmitted and confine the light to the core.

- An optical fiber consists of a light-carrying core in the center surrounded by a cladding that acts to trap the light in the core. Glass fiber is covered by a plastic buffer/jacket coating that protects it from the environment and allows easy handling for splicing or termination.

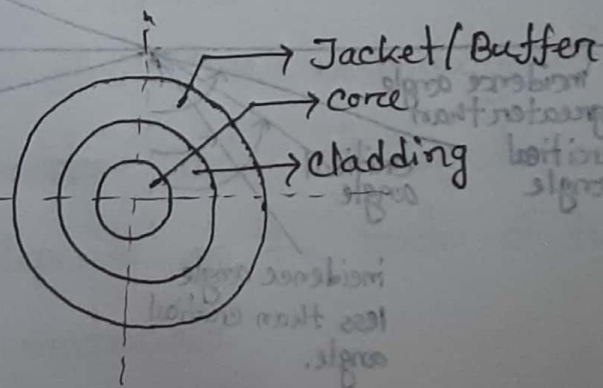


Figure: Transverse view

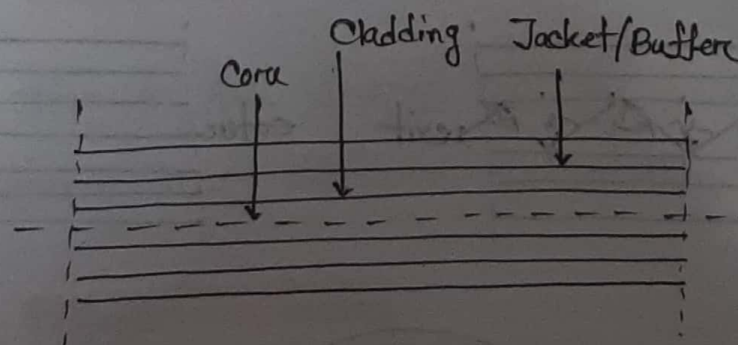


Figure: Longitudinal view

* Types of Optical Fiber cables:

- i) Single mode fiber
- ii) Multimode fiber
- iii) Step-index fiber
- iv) Graded index fiber

* Single Mode Vs Multi mode

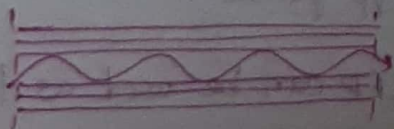
Single Mode fiber	Multi Mode fiber
i) Very small core diameters; a few μm .	i) Much larger core diameters; tens to hundreds μm .
ii) Launching of optical power is difficult	ii) Launching of optical power is easy.
iii) no intermodal dispersion	iii) intermodal dispersion occurs.
iv) Propagation of only one mode	iv) Many modes propagate at same time.
v) Duration of input, output pulse is same	v) o/p pulse is spread out compared to i/p pulse.
vi) LED can't be used as light source.	vi) LED can be used as light source
vii) Higher bandwidth.	vii) Less bandwidth
viii) Applicable for long distance communication.	viii) Applicable for short distance communication.

* Step index fibre vs Graded index fibre.

Step index fibre	Graded index fibre
i) RI of core is uniform.	i) Refractive index ^{of core} is not uniform
ii) Pulse distortion is present	ii) Pulse distortion is minimum
iii) Can be single mode or multimode	iii) Only multimode
iv) Can be manufactured easily	iv) Manufacturing is not easy
v) High numerical aperture	v) Low numerical aperture
vi) Attenuation is higher	vi) Attenuation is lower.
vii) Offers lower bandwidth	vii) Offers higher bandwidth
viii) Reflection losses are present	viii) Reflection losses are absent

Graded index fibre: A cable whose core RI is non uniform and varies gradually. RI is highest in the center of core and gradually diminishes towards cladding.

Cladding
Core
Cladding

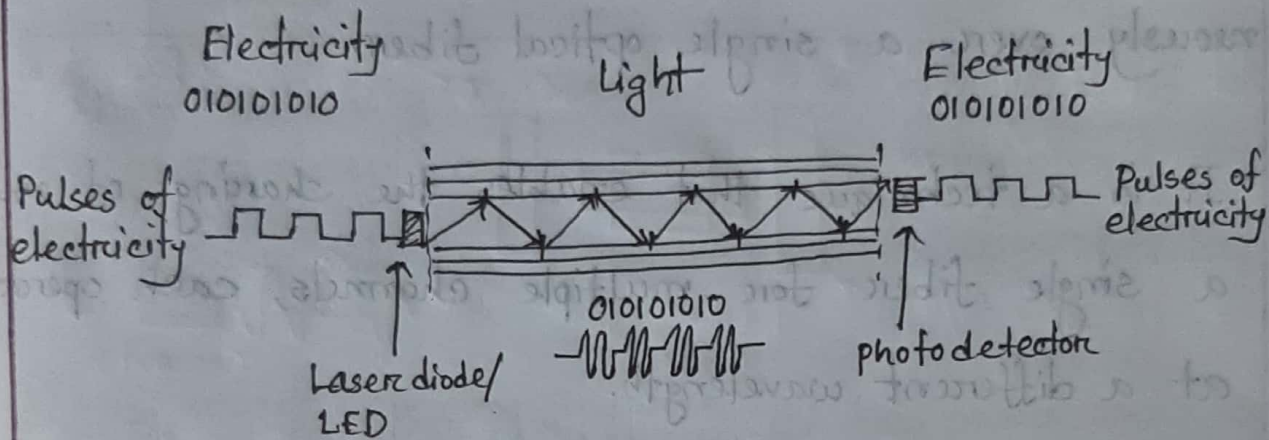


Step index fibre: A fiber optic cable with a uniform refractive index throughout the core.

Cladding
Core
Cladding



* Fiber optics Communication System *



- Pulses of electricity corresponding to the bits arrive at the input transducer (A device that converts one form of energy to another).
- LED/Laser diodes are used as optical transducers at the input of fiber optic cables to convert electricity into lights.
- Electrical signals arriving at the input of optoelectronic devices are used to modulate the light source.
- The modulated optical signal is emitted by the source and coupled into the cable.
- Once the light is trapped inside the cable, it travels to the other end where it is demodulated and an output transducer such as photodiode or photo-transistor is used to convert the demodulated lights into electrical pulses.

* **Wavelength Division Multiplexing**: is a technology used in fiber optics to transmit multiple optical signals simultaneously over a single optical fiber.

- It is a technique that enables the sharing of multiple a single fiber for multiple channels, each operating at a different wavelength.

- increased capacity and efficiency in of optical communication systems.

- By utilizing different wavelengths, multiple independent data streams can be combined and transmitted over a single fiber without interfering with each other.

- The basic principle of WDM involves the use of optical filters to separate and combine different wavelength of light.

- Two main types of WDM are:

i) Coarse wavelength division multiplexing (CWDM)

ii) Dense wavelength division multiplexing (DWDM)

* CWDM vs DWDM

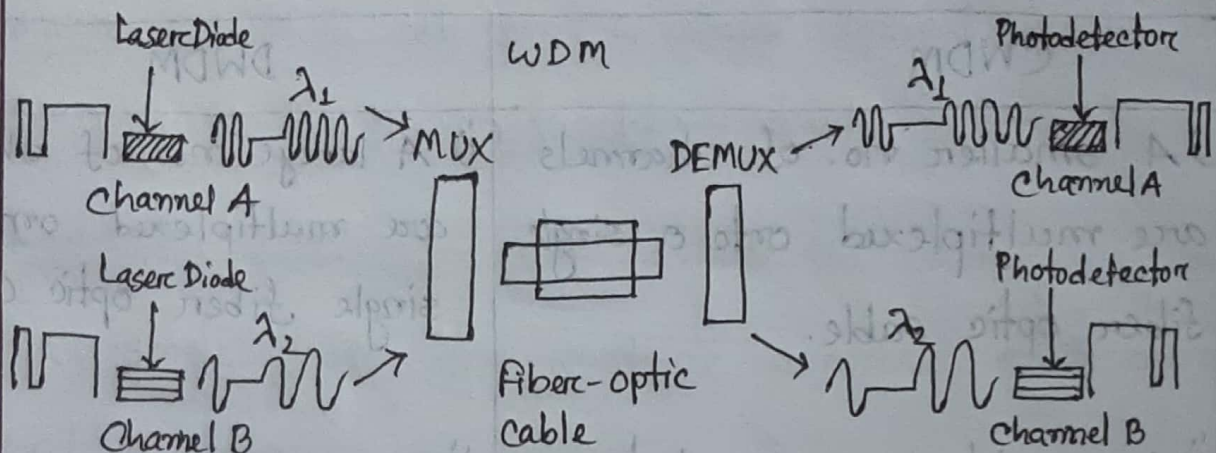
CWDM

- i) A smaller no. of channels are multiplexed onto a single fiber optic cable.
- ii) has wider spacing between channels
- iii) Supports a smaller no. of channels, usually upto 18.
- iv) Commonly used for shorter distance applications.
- v) Typically have lower data rates
- vi) lower spectral efficiency
- vii) Less expensive

DWDM

- i) A large no. of channels are multiplexed onto a single fiber optic cable
- ii) has narrower spacing of channels.
- iii) Supports significantly higher no. of channels.
- iv) It is preferred for long-haul and ultra-long-haul applications.
- v) Support higher data rates
- vi) Greater spectral efficiency
- vii) More expensive.

**** A fiber-optic communication system employing WDM:**



How WDM works:

1. Source signal - multiple data streams or sources are generated, each representing different information to be transmitted.
2. Wavelength selection - each data stream is assigned a specific wavelength of light. The wavelengths are carefully chosen to ensure they are separable and do not interfere with e/o.
3. Wavelength combining - The individual data streams are combined using an optical multiplexer into a single optical signal for transmission.
4. Transmission - The combined optical signal, carrying is launched into the optical fiber transmission. The signal travels along the fiber, maintaining its wavelength characteristics.
5. Wavelength separation - At the receiving end, an optical demultiplexer is used to separate the combined signal into

Protection and Restoration

its constituent wavelengths. This device acts as a filter and directs each wavelength to the appropriate receiver or detector.

6. Detection and Processing - The individual data streams are received by their respective detectors. These detectors convert the optical signals back into their original electrical form for further processing and decoding.

* Fiber Optic Technology **benefits:**

- i) High transmission rate
- ii) High security
- iii) Small weight and size
- iv) Low power consumption
- v) Immunity to Electromagnetic Interference
- vi) Low attenuation

* **Drawbacks:**

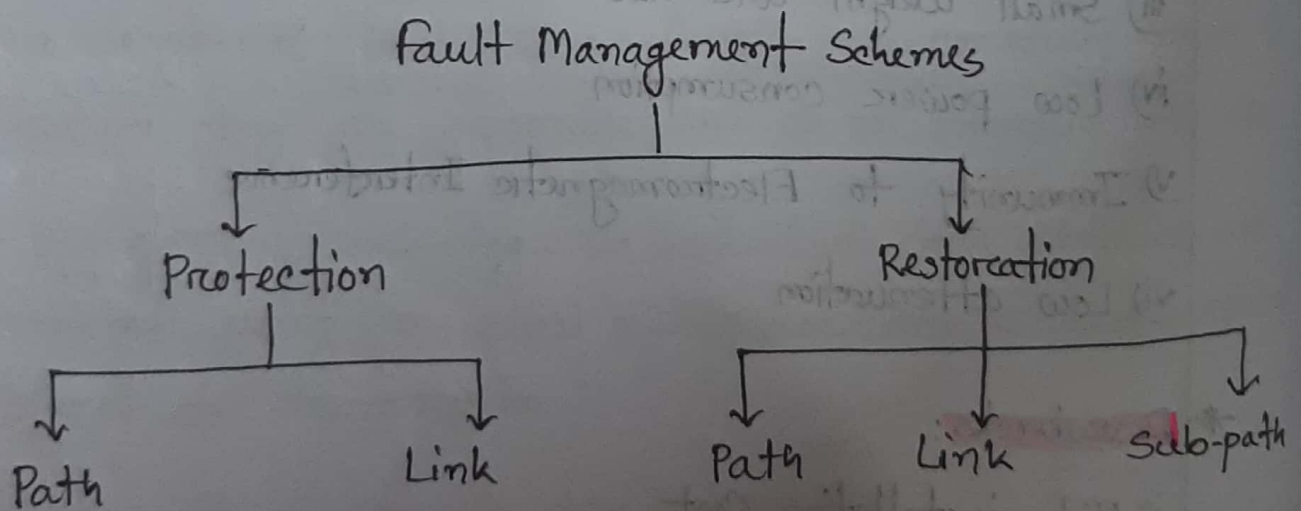
- i) High installation cost
- ii) Difficulty in splicing
- iii) High maintenance cost

Protection and Restoration

* In a WDM Network, failure of a network component may cause failure of several optical channels leading to large data loss which can interrupt communication services.

* **Fault Management Schemes:**

Fault management in optical network is done through reserving backup resources in advance called protection or discovering spare backup resources in an online manner called restoration.



* **Link Protection:** Backup paths and wavelengths are reserved around each link on the primary path during connection setup. When a link fails, all the connections traversing the failed link will be rerouted.

around that link and the source and destination nodes of the connections traversing the failed link would be unaffected by the link failure.

* **Path Protection Restoration**: Source and destination of each connection statically reserve a primary path and a backup path on an end-to-end basis during connection setup. When a link fails, the source and destination node of each connection that traverses the failed link are informed about the failure and backup resources are utilized.

* **Link Restoration**: In the event of a failure, the end nodes of the failed link participate in a distributed algorithm to dynamically discover a new route around the link. When a new link is route is discovered around the failed link for a wavelength channel, the end nodes of the failed link reconfigure their optical interconnections to reroute that channel onto the new route.

* **Path Restoration**: The source and destination nodes of each connection independently discover a backup route on an end-to-end basis and such a backup path can be on a different wavelength channel. When a new route and wavelength channel is discovered for a connection, network elements are configured appropriately and the connection switches to the new path.

* **Sub-path Restoration**: When a link fails, the upstream node of the failed link detects the failure and discovers a backup route from itself to the corresponding destination node for each disrupted connection. Upon successful discovery of resources for the new backup route, intermediate optical interconnections are reconfigured appropriately and the connection switches to the new path.

Optical Connectors

* **Optical Connector**: An optical fiber connector joins optical fibers and enables quicker connection and disconnection than splicing. The connectors mechanically couple and align the cores of fibers so light can pass.

* **Principle requirements of a good connector design**:

1. **Low coupling losses**: the connector assembly must ~~not~~ ~~change~~ ~~str~~ maintain strict alignment tolerance to ensure low loss.
2. **Interchangeability**: connectors of the same type must be compatible from one manufacturer to another.
3. **Ease of assembly**: A service technician should be able to install the connector easily in field environment.
4. **Low environmental sensitivity**: conditions such as temperature, dust, moisture etc should have a small effect.
5. **Low cost & reliable construction**: cost must not be a major factor in the fiber system.
6. **Ease of connection**: one should be able to ~~mad~~ mate & demate the connector simply by hand.

Optical Connectors

* Connector Types:

- most used: snap-on & twist-on design.

- Basic coupling mechanism:

i) Butt-joint connector

ii) Expanded-beam connector

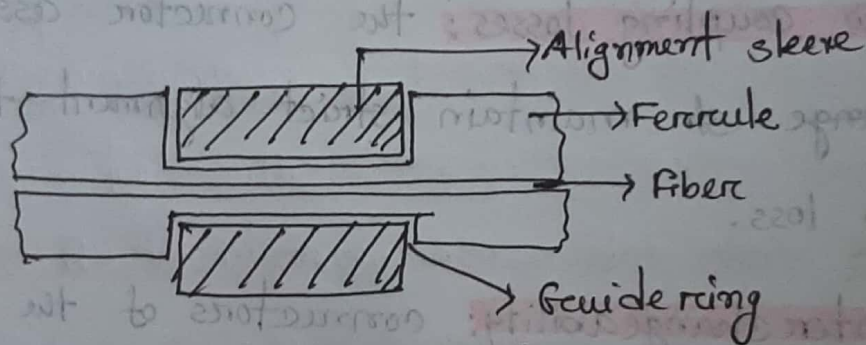


Fig: An alignment scheme for butt joint

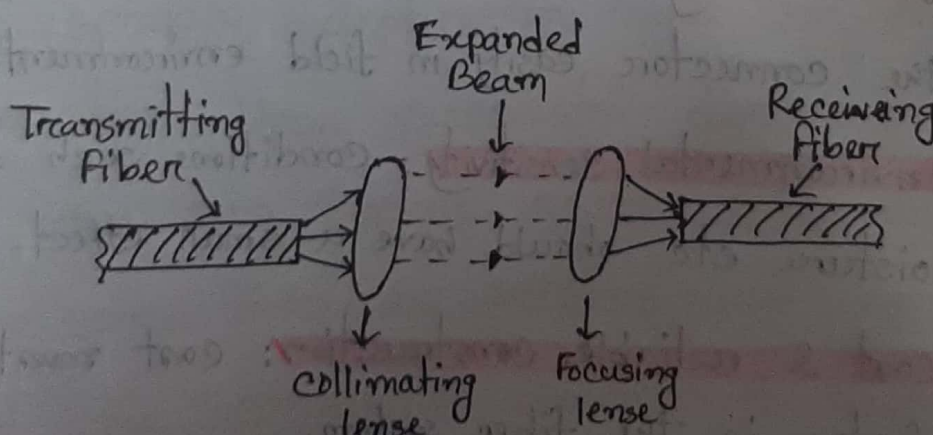


Fig: Expanded-beam connector

* Six popular fiber optic connectors:

1. **ST** (Straight tip) connector

- uses ceramic ferrule and rugged metal housing
- latched in place by twisting
- Typical loss range 0.20 - 0.50 dB
- Designed for distribution applications using either single mode or multimode fibers.

2. **SC** (Subscriber/Square Connector)

- uses ceramic ferrule in simplex or duplex plastic housing
- snap-in connection for tight spaces
- Typical loss range 0.20 - 0.45 dB
- widely used in Gigabit ethernet, LAN, MAN, WAN, ATM

3. **LC** Connector

- uses a standard RJ-45 telephone plug housing and ceramic ferrules in simplex or duplex plastic housing.
- Typical loss range 0.10 - 0.50 dB
- Available in simplex or duplex configuration for CATV, LAN, MAN, WAN.

4. **MU** (Miniature Unit) Connector

- SFF connector based on a 1.25mm ceramic ferrule and a single free floating ferrule.
- Typical loss range 0.10 - 0.30 dB
- Used mainly in Japan; Suitable for board-mounted applications

5. **MT-RJ** (Media Termination - Recommended Jack) Connector

- SFF connector with 2 fibers in one moulded plastic ferrule and an improved RJ-45 latch mechanism
- Typical loss range 0.25 - 0.75 dB
- Applications are for MANs and LANs

6. **MPO** (Multiple fiber push on/pull off) MTP connector

- Can house upto 12 multimode or single mode optical fibers in a single compact ferrule.
- Typical loss range 0.25 - 1.00 dB
- Allows high density connection between network equipment in telecom rooms.